ON TWO OPEN QUESTIONS OF GAISER ON RESTRICTED GENERALIZED SCHUR NUMBERS

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ABSTRACT. For integers $2 \le \ell \le k$, let $S_2(k;\ell)$ be the least n such that every red–blue colouring of $\{1,\dots,n\}$ contains a monochromatic solution of $x_1+\cdots+x_k = x_{k+1}$ that uses exactly $\ell+1$ distinct integers. Gaiser proved that $S_2(k;\ell) \ge F(k,\ell) := k^2 + \left[\frac{(\ell+1)(\ell-2)}{2} + 2\right]k + \ell(\ell-2)$ for all $2 \le \ell \le k$, with equality for all sufficiently large k, and asked for the smallest threshold $K(\ell)$ from which equality holds (Open Question 6.4). He also asked whether the three-colour analogue satisfies $S_3(k;2) = k^3 + 3k^2 + k - 1$ for large k (Open Question 6.2). We show that $S_3(k;2) = k^3 + 3k^2 + 2k - 2$ for every $k \ge 3$, so the answer to Open Question 6.2 is negative; the lower bound had been recorded publicly before, to the best of our knowledge the matching upper bound has not been established before; we certify it by polynomial hypergraph certificates, LRAT proofs and a Lean 4 formalisation. We give computer-assisted proofs that $K(3) = 3$, $K(4) = 4$, $K(5) = 6$, $K(6) = 7$, $K(7) = 8$, $K(8) = 9$, $K(9) = 10$, $K(10) = 11$ and $K(11) = 12$. The upper bounds hold for *all* k beyond a fixed k_0 simultaneously; they are certified by finite hypergraphs whose vertices are integer polynomials in k, together with LRAT proofs of unsatisfiability. For $\ell = 8, 9, 10, 11$ one remaining value of k is settled by an ordinary (numerical) hypergraph certificate; for $\ell = 11$ the symbolic part ($k \ge 13$) consists of 222 certificates, one for each cube of a complete partition of the colourings by prefixes. All LRAT proofs were accepted by two checkers, one of them formally verified. For $\ell \le 7$ the certificates were also checked by a second, independently written implementation, and the argument is formalised in Lean 4; the cases $\ell = 8, 9, 10$ have so far been checked only with first-party tools (written and run within this project); for $\ell = 11$ all 222 symbolic certificates for $k \ge 13$ were also replayed by a second, independently written implementation, run by a second party. The failure of the formula at $k = \ell$ for $\ell \ge 5$ follows from a known lower bound for weak Schur numbers. In the other direction, the threshold grows linearly. New in v1.2: a three-block colouring together with a new forcing template for windows with a sparse colour gives $(1 + a_*)\ell - 4 < K(\ell) < (1 + a_*)\ell + 5$ for every $\ell \ge 50$, where $a_* = (\sqrt{2} - 1)/2$. Hence $K(\ell) = \frac{1+\sqrt{2}}{2}\ell + O(1)$. The colouring also gives $K(19) \ge 21$ and $K(20) \ge 22$. (Version 1.1 had four-block colourings giving $K(\ell) \ge \ell + 2$ for $\ell \ge 24$ and $K(\ell) \ge (\frac{9}{8} - o(1))\ell$.) For $k = \ell$ the three-block colourings exceed the known lower bound for two-colour weak Schur numbers when $k \ge 20$, for example $WS_{20}(2) \ge 4770$; the leading term is still $(\frac{9}{16} - o(1))k^3$. Our certificates show that the earlier lower bound of Ahmed, Boza, Revuelta and Sanz is attained for $k = 8, 9, 10$. New in v1.3: by an exact form of the forcing template, a reduction lemma and 14,723 per-window SAT certificates, $K(\ell) = \ell+1$ for every $12 \le \ell \le 18$, $K(19) = 21$ and $K(20) = 22$, and $S_2(20;19) = 4181$, $S_2(21;20) = 4861$; these are the two smallest pairs with $k > \ell$ and $S_2 > F$. A prefix-dominance lemma for the forcing template gives $(1 + a_*)\ell - 3.63 < K(\ell) \le (1 + a_*)\ell + 16 - 11\sqrt{2}$ for every $\ell \ge 19$.

Date: Preprint v1.3, 8 November 2026 (v1.2: 3 November 2026; v1.1: 1 November 2026; v1.0: 28 October 2026). Computer-assisted proof candidate; not peer reviewed.

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Status of this preprint. Computer-assisted proof candidate; not peer reviewed. For $\ell \le 7$ the certificate chain has been replayed by a second, independently written checker (§4); the Lean formalisation has so far been run only within this project. The cases $\ell = 8, 9, 10$ (Theorem 3(c)) are newer: they have been checked only with first-party tools (written and run within this project) and are not formalised. The case $\ell = 11$ (Theorem 3(d), new in v1.1, §4.1): its 222 symbolic certificates for $k \ge 13$ were checked with first-party tools and replayed in full by a second, independently written implementation run by a second party; its $k = 12$ certificate has been checked only with first-party tools; it is not formalised. For Theorem 30 (Open Question 6.2) all certificates have been replayed by a second, independently written checker; its Lean formalisation has been run only within this project. Theorem 5 and Corollary 6 (23 October) are elementary; their proofs, including the asymptotics and the sharpness remark, have been checked by a second party on paper and with a computer algebra system, but their Lean formalisation has been run only within this project. The literature check for Corollary 6(c) and for the new values of $WS_k(2)$ covers the published sources we could access; one announced follow-up paper of Ahmed et al. could not be obtained. "Independently written" refers to the implementation of a checker; a second-party replay means that a certificate was actually executed and checked by a separate system. *New in v1.2:* Theorem 18 (§8). Its proofs are written out; the exact symbolic and finite checks it uses were run with two separately written sets of programs, one by each AI system. A second party (GPT) reviewed the section, including Proposition 15 and Lemma 16 on which the upper bound depends. It is not formalised. *New in v1.3:* Theorem 25, Lemma 28 and Theorem 29 (§8.1, §8.2). The computations and the 14,723 per-window certificates are first-party. A second party (GPT) reviewed the written proofs, including Lemma 28 and Theorem 29, with its own symbolic and finite-window checks, re-enumerated the windows for $\ell = 12, 13$ and the reduced windows for $\ell \le 20$ with its own programs, and wrote an independent CNF builder and LRAT checker for the certificates; that checker was executed by Claude on all 14,723 certificates (all accepted), not by GPT, because the data could not be transferred to it. The second party's verdict (8 November 2026) was ACCEPT for Theorem 25, Lemma 28 and Theorem 29 at this level; Lemma 28 was reviewed as a written proof, with symbolic and finite-window machine checks. The Lean module `OQ64.Three` has been run only within this project. The human author has not personally verified every proof step. Extensive use of AI systems is described in Appendix A.

1. Introduction

Throughout, a *solution* is a tuple $(x_1, \dots, x_{k+1})$ of positive integers with $x_1 + \cdots + x_k = x_{k+1}$, and $c(S)$ denotes the number of distinct integers among its entries. A colouring is *good* for (k, ℓ, n) if it is a red–blue colouring of $[1, n] = \{1, \dots, n\}$ with no monochromatic solution S in $[1, n]$ with $c(S) = \ell + 1$.

Definition 1 (Gaiser [11])**.** For $2 \le \ell \le k$, $S_2(k; \ell)$ is the least n such that no colouring is good for (k, ℓ, n).

Write

$$F(k,\ell) = k^2 + \Big[\frac{(\ell+1)(\ell-2)}{2} + 2\Big]k + \ell(\ell-2), \qquad D = \binom{\ell}{2}, \qquad E = \binom{\ell-1}{2}.$$

Theorem 2 (Gaiser [11, Thm. 1.1])**.** *For all $2 \le \ell \le k$, $S_2(k; \ell) \ge F(k, \ell)$. For each fixed $\ell \ge 2$, equality holds for all sufficiently large k.*

Section 9 treats Gaiser's Open Question 6.2 (three colours, $\ell = 2$); the rest of the paper is about Open Question 6.4.
Gaiser's Open Question 6.4 asks, for $\ell \ge 3$, for the smallest K such that $S_2(k; \ell) = F(k, \ell)$ for all $k \ge K$. We denote it by $K(\ell)$; note $K(\ell) \ge \ell$ by definition.

Theorem 3 (main result; computer-assisted)**.**
(a) $S_2(k; 3) = (k+1)(k+3)$ *for all* $k \ge 3$, *and* $S_2(k; 4) = k^2 + 7k + 8$ *for all* $k \ge 4$.
(b) $S_2(k; 5) = k^2 + 11k + 15$ *for all* $k \ge 6$, $S_2(k; 6) = k^2 + 16k + 24$ *for all* $k \ge 7$, *and* $S_2(k; 7) = k^2 + 22k + 35$ *for all* $k \ge 8$.

(c) $S_2(k;8) = k^2 + 29k + 48$ *for all* $k \geq 9$, $S_2(k;9) = k^2 + 37k + 63$ *for all* $k \geq 10$, *and* $S_2(k;10) = k^2 + 46k + 80$ *for all* $k \geq 11$.
(d) $S_2(k;11) = k^2 + 56k + 99$ *for all* $k \geq 12$.
(e) *Consequently* $K(3) = 3$, $K(4) = 4$, $K(5) = 6$, $K(6) = 7$, $K(7) = 8$, $K(8) = 9$, $K(9) = 10$, $K(10) = 11$ *and* $K(11) = 12$.

The lower bounds in (a)–(d) are Theorem 2. The new content is the upper bounds, each valid for *every* k from the stated threshold on. Part (e) also uses the known fact that the formula fails at $k = \ell$ for $\ell \geq 5$ (Proposition 4 below).
Theorem 3 might suggest that $K(\ell) = \ell + 1$ for every $\ell \geq 5$ (an earlier version of this draft conjectured exactly this). This is false: by Corollary 6 below, $K(\ell) \geq \ell + 2$ for all $\ell \geq 24$, and by Theorem 18 (new in v1.2) $K(\ell) = \frac{1+\sqrt{2}}{2}\ell + O(1)$, with $K(19) \geq 21$. New in v1.3, Theorem 25 shows that $K(\ell) = \ell + 1$ for $12 \leq \ell \leq 18$, $K(19) = 21$ and $K(20) = 22$; so the first failure at $k = \ell + 1$ occurs exactly at $\ell = 19$.

2. Prior work and lower bounds

Lower bound. Gaiser's lower bound uses the three-block colouring red on $[1, s-1]$, blue on $[s, t-1]$, red on $[t, F-1]$, where $s = k + D$ and $t = ks + D$. Note that $F(k, \ell) = t + k - 1 + E$.

The diagonal $k = \ell$. When $k = \ell$, a solution has exactly $\ell + 1$ distinct entries precisely when all its entries are distinct, so $S_2(k;k) = S_2^*(k)$, the two-colour generalized *weak* Schur number [11]. In the notation $WS_k(2)$ of the weak Schur literature (with k summands), the known values are

$$WS_2(2), \ldots, WS_7(2) = 9,\ 24,\ 52,\ 101,\ 166,\ 253.$$

The value $WS_2(2) = 9$ appears as $h(2) = 9$ in Blanchard, Harary and Reis [4]. For the equation $x_1 + \cdots + x_k + c = x_{k+1}$ with distinct summands, Boza, Marín, Revuelta and Sanz [3] proved $WR_3(2, c) = 5c + 24$ for every even $c \geq 0$ ($WR_3(2, c) = \infty$ for odd c) and $WR_4(2, c) = 6c + 52$ for every $c \geq 0$; $c = 0$ gives $WS_3(2) = 24$ and $WS_4(2) = 52$. Ahmed et al. [1] cite both [3] and Sanz's thesis [6] for the early results on $k = 3, 4$; we have not inspected the thesis. $WS_5(2) = 101$ is due to [3]; see also [5]. $WS_6(2) = 166$ and $WS_7(2) = 253$ are Theorems 4.2 and 4.6 of Ahmed, Boza, Revuelta and Sanz [1].

Proposition 4 (Ahmed–Boza–Revuelta–Sanz [1, Lemma 2.1])**.** *For* $k \geq 5$, $WS_k(2) \geq \frac{k^3+4k^2-5k+2}{2}$. *The colouring* $A_1 = \{1\} \cup \left[\frac{k^2+3k}{2}, \frac{k^3+3k^2-6k+2}{2}\right]$, $A_2 = \left[2, \frac{k^2+3k-2}{2}\right] \cup \left[\frac{k^3+3k^2-6k+4}{2}, \frac{k^3+4k^2-5k}{2}\right]$ *has no monochromatic solution with distinct entries.*

Since $F(\ell, \ell) = \frac{\ell^3+3\ell^2-2\ell}{2}$, the bound equals $F(\ell, \ell) + E$ at $k = \ell$, and $E > 0$ for $\ell \geq 3$. Hence $S_2(\ell;\ell) > F(\ell, \ell)$ and $K(\ell) \geq \ell + 1$ for all $\ell \geq 5$; this is an immediate consequence of known results, and we make no claim to it. For $\ell = 3, 4$ the known values $WS_3(2) = 24 = F(3, 3)$ and $WS_4(2) = 52 = F(4, 4)$ agree with Theorem 3(a).
New values. With the method of Lemma 10 we also obtained $WS_8(2) = 365$, $WS_9(2) = 505$ and $WS_{10}(2) = 676$. The upper bounds are certified by hypergraphs with LRAT proofs (accepted by `lrat-check` and `cake_lpr`); the lower bounds are Proposition 4. Our search, without certificates, also gives $WS_{11}(2) = 881$, $WS_{12}(2) = 1123$, $WS_{13}(2) = 1405$ and $WS_{14}(2) = 1730$. So the bound of Proposition 4 is attained for $5 \leq k \leq 10$, and our uncertified computations support equality also for $11 \leq k \leq 14$; by Corollary 6(c) below it is not attained for $k \geq 22$. In the published sources we could check (including [2]), $WS_k(2)$ is known exactly only for $k \leq 7$; as of 28 October 2026, OEIS A365190 lists the values at $k = 8, 9, 10$ only as lower bounds, and we have not located a published proof of the matching upper bounds. This limited literature search does not by itself establish priority.

Four-block colourings with a short blue block. The colouring in Proposition 4 has the shape: red $\{1\}$, blue $[2, j+1]$, red $[j+2, m]$, blue $[m+1, M]$, with $j \approx k^2/2$. Taking j much smaller, about $\ell/4$, gives better colourings when ℓ is large, and the same works for $k > \ell$.

Theorem 5. *Let $k \geq \ell \geq 2$, $e = k - \ell$ and $1 \leq j \leq \ell - 1$. Put*

$$m = e + (\ell - 1)j + \frac{\ell(\ell+1)}{2} - 1, \qquad V = \frac{j(j+3)}{2} + (\ell - j)\, m + \frac{(\ell - j)(\ell - j + 1)}{2} + 2e.$$

The colouring of $[1, V-1]$ with red class $\{1\} \cup [j+2, m]$ and blue class $[2, j+1] \cup [m+1, V-1]$ is good for $(k, \ell, V-1)$. Hence $S_2(k; \ell) \geq V$.

Proof. In a monochromatic solution the sum x_{k+1} has the colour of the summands, and the k summands take exactly ℓ distinct values. Let S be a colour class with elements $s_1 < s_2 < \cdots$; if $|S| < \ell$ there is nothing to show. If the summands take the values $v_1 < \cdots < v_\ell$ in S with multiplicities $\mu_i \geq 1$, then $v_i \geq s_i$ and $\sum_i \mu_i v_i = \sum_i v_i + \sum_i (\mu_i - 1) v_i \geq s_1 + \cdots + s_\ell + (k - \ell) s_1$. For the red class this bound is $1 + \sum_{i=2}^{\ell} (j + i) + (k - \ell) = m + 1$. The blue class has only $j < \ell$ elements below $m+1$, so its bound is $\sum_{i=2}^{j+1} i + \sum_{i=1}^{\ell-j} (m + i) + 2(k - \ell) = V$. In both cases the sum exceeds every element of the class, so no monochromatic solution exists. □

Corollary 6.
- *(a) For every $\ell \geq 24$, $S_2(\ell+1; \ell) > F(\ell+1, \ell)$, hence $K(\ell) \geq \ell + 2$. For example $S_2(25; 24) \geq 8097 > 8078 = F(25, 24)$ and $S_2(26; 25) \geq 9132 > 9077 = F(26, 25)$.*
- *(b) For $\ell \geq 26$ and every integer e with $1 \leq e \leq (\ell - 18)/8$, $S_2(\ell + e; \ell) > F(\ell + e, \ell)$. Hence $K(\ell) \geq \ell + 1 + \lfloor (\ell - 18)/8 \rfloor$ and $K(\ell) \geq (\frac{9}{8} - o(1))\ell$.*
- *(c) For every $k \geq 22$, $WS_k(2) = S_2(k; k)$ exceeds the bound $\frac{k^3 + 4k^2 - 5k + 2}{2}$ of Proposition 4; asymptotically $WS_k(2) \geq (\frac{9}{16} - o(1))k^3$.*

Proof. Let $r \in \{0, 1, 2, 3\}$ with $r \equiv \ell \pmod 4$, take $j = (\ell - r)/4$ in Theorem 5, and put $G = V - F(\ell + e, \ell)$. A direct expansion gives

$$G = \frac{\ell^3}{16} - \Big(\frac{e}{2} + 1\Big)\ell^2 + \Big(\frac{16 + r(6 - r)}{16} - \frac{3e}{4}\Big)\ell - e^2 + \Big(1 + \frac{r}{4}\Big)e - \frac{r(4 - r)}{8}.$$

(b) $\partial G/\partial e = -2e - \frac{\ell^2}{2} - \frac{3\ell}{4} + 1 + \frac{r}{4} < 0$ for $e \geq 0$, so it suffices to take $e = (\ell - 18)/8$. There $G = \frac{\ell^2}{64} + \alpha_r \ell - \beta_r$ with $\alpha_r \in \{\frac{27}{8}, \frac{119}{32}, \frac{63}{16}, \frac{129}{32}\}$ and $\beta_r \in \{\frac{117}{16}, \frac{33}{4}, \frac{143}{16}, \frac{75}{8}\}$, which is positive for $\ell \geq 3$. (a) For $\ell \geq 26$ take $e = 1$ in (b). For $\ell = 24, 25$ take $j = 5$; the values are as stated. (c) For $e = 0$, $V - \frac{k^3 + 4k^2 - 5k + 2}{2} = G - \binom{k-1}{2} = \frac{k^3}{16} - \frac{3}{2}k^2 + \big(\frac{16 + r(6-r)}{16} + \frac{3}{2}\big)k - 1 - \frac{r(4-r)}{8}$ (with $\ell = k$), which is positive for $k \geq 24$; for $k = 22, 23$ take $j = 5$ ($V = 6242 > 6238$ and $V = 7121 > 7085$). The asymptotic statement is $V = \frac{9}{16}k^3 + O(k^2)$. □

The four-block colourings themselves were first found by our search program for $(k, \ell) = (26, 25)$ and $(31, 30)$; Theorem 5 was then written down by hand. As a check, a separately written exact program (a bit-set dynamic programme over the number of summands and of distinct values) confirmed that the colourings in Theorem 5 with the optimal j are good for ten pairs (k, ℓ) between $(22, 22)$ and $(42, 40)$ (and, for $(26, 25)$, also with $j = 6$). Theorem 5 itself, for all k, ℓ, j, is also formalised in Lean 4 (module `OQ64.Four`), together with the instances $S_2(25; 24) > F(25, 24)$, $S_2(26; 25) > F(26, 25)$ and $S_2(22; 22) \geq 6242$. The polynomial inequalities behind Corollary 6 are not formalised; they were checked with a computer algebra system. For $\ell \leq 23$, no choice of j beats $F(\ell + 1, \ell)$. Corollary 6(b) is sharp for this family when $\ell \geq 74$: maximising G over real j gives $G_{\max}(e)$, which is decreasing in e and equals $-\frac{12\ell^3 - 1025\ell^2 + 4222\ell - 4489}{256(\ell - 2)} < 0$ at $e = (\ell - 17)/8$ for $\ell \geq 82$ (the cases $74 \leq \ell \leq 81$ were checked directly). So for $\ell \geq 74$ the largest k for which some colouring of Theorem 5 beats $F(k, \ell)$ is exactly $\ell + \lfloor (\ell - 18)/8 \rfloor$. For $5 \leq k \leq 21$ the best choice is the large j of Proposition 4, and the two bounds coincide. We do not know whether Corollary 6(c) is new: we found no improvement of Proposition 4 in the literature we checked, but that check is

not complete (the brief announcement [2] does not contain it; the authors' follow-up paper, listed as submitted, could not be checked).

3. Symbolic hypergraph certificates

Fix ℓ and $k_0 \geq \ell$. All polynomials below are integer polynomials of degree at most 2 in k, and "for all k" means for all integers $k \geq k_0$.

Definition 7. A *certificate* for (ℓ, k_0) consists of

(i) a finite list of polynomials $p_1, \ldots, p_N$ (*vertices*) with $p_1 = 1$, such that $1 \leq p_i(k) \leq F(k, \ell)$ for all k;

(ii) a finite list of *edges*. An edge is $(a; c; (m_1, o_1), \ldots, (m_{\ell-1}, o_{\ell-1}); w)$, where a, o_j, w are vertices, $\deg a \leq 1$, the m_j are positive integers with $\sum_j m_j = c \leq k_0 - 1$, and

$$w = (k - c)\, a + \sum_{j=1}^{\ell-1} m_j\, o_j \quad \text{as polynomials,}$$

and, for all k, the ℓ summand values $a(k), o_1(k), \ldots, o_{\ell-1}(k)$ are pairwise distinct;

(iii) the CNF formula over one Boolean variable per vertex (true = red) consisting of the unit clause $\{p_1\}$ and, for each edge, the two clauses "not all of $a, o_1, \ldots, o_{\ell-1}, w$ are red" and "not all of them are blue". This CNF is unsatisfiable.

Lemma 8 (soundness)**.** *If a certificate for (ℓ, k_0) exists, then $S_2(k; \ell) \leq F(k, \ell)$ for every $k \geq k_0$.*

Proof. Fix $k \geq k_0$ and a colouring χ of $[1, F(k, \ell)]$. Swapping colours if necessary, $\chi(1)$ is red. Give each vertex p_i the colour $\chi(p_i(k))$, which is defined by (i). This is an assignment of the CNF that satisfies the unit clause, so by (iii) it falsifies the clause of some edge. Thus $a(k)$, the $o_j(k)$ and $w(k)$ all have the same colour. The summand $a(k)$ is taken with multiplicity $k - c \geq 1$ and $o_j(k)$ with multiplicity $m_j \geq 1$; these multiplicities add up to k, so the identity in (ii) is a solution with k summands. Its summands take exactly ℓ distinct values by (ii), and $w(k)$ exceeds each of them because it is a sum of at least two positive terms that include that summand. So the solution has exactly $\ell + 1$ distinct integers, all in $[1, F(k, \ell)]$, and it is monochromatic. □

Remark 9. Two different vertices may take the same value at some k. This does not affect the proof: the assignment induced by χ is still *an* assignment, and every assignment falsifies some clause. Using separate variables only weakens the constraints.

Checking the conditions exactly. Each condition in (i) and (ii) has the form $q(k) \geq 0$ for all integers $k \geq k_0$, with $\deg q \leq 2$. Substituting $k = k_0 + t$, it suffices to evaluate q at $t = 0$ and at the integers next to the real vertex of the parabola, when the leading coefficient is positive. A negative leading coefficient, or a degree-one polynomial with negative slope, fails. No sampling is involved.

Finding certificates. The certificates were found by a DPLL-style search. It branches on the colours of the integers $2, 3, \ldots$, and in each branch it propagates forced colours through identities of the shape in (ii) among same-coloured vertices. New vertices are created as needed. The identities used in contradictions are recorded and then reduced to a small unsatisfiable core with a SAT solver. For $\ell = 8, 9, 10$ the search only uses identities with $c \in \{\ell - 1, \ell\}$, which keeps the number of candidate identities manageable while condition (ii) ($c \leq k_0 - 1$) still holds. The search procedure plays no role in the correctness argument.

Single values of k. For one fixed k we use ordinary hypergraphs on integers.

Lemma 10. *Let $k \geq \ell \geq 2$ and $n \geq 1$. Let H be a finite list of solutions $x_1 + \cdots + x_k = x_{k+1}$ in $[1, n]$, each with exactly $\ell + 1$ distinct integers, and suppose the CNF over variables $1, \ldots, n$ consisting of the unit clause $\{1\}$ and, for each member of H, the clauses "not all of its integers are red" and "not all of them are blue" is unsatisfiable. Then $S_2(k; \ell) \leq n$.*

Proof. A colouring of $[1, n]$ with 1 red is an assignment satisfying the unit clause, so it falsifies a clause of some member of H, which is then monochromatic. If 1 is blue, swap the colours first. □

The lists H were found by a DPLL search over the colours of $2, 3, \ldots,$ in which forced colours are propagated by a dynamic programme over the integers of each colour class; the solutions that justify forcings and contradictions are recorded and reduced to an unsatisfiable core.

4. The certificates and their verification

ℓ	k_0	$F(k,\ell)$	vertices	edges	CNF clauses	LRAT additions
3	3	k^2+4k+3	67	103	207	92
4	4	k^2+7k+8	181	387	775	412
5	6	$k^2+11k+15$	288	836	1673	623
6	7	$k^2+16k+24$	864	3298	6597	2318
7	8	$k^2+22k+35$	843	11168	22337	7858
8	10	$k^2+29k+48$	1538	42170	84341	61589
9	11	$k^2+37k+63$	2707	160068	320137	183492
10	12	$k^2+46k+80$	4448	609886	1219773	662622

Table 1. The single-family symbolic certificates for Theorem 3 with $3 \le \ell \le 10$. Each proves $S_2(k;\ell) \le F(k,\ell)$ for all $k \ge k_0$. The $\ell = 11$ symbolic upper bound is certified instead by the cube family of Table 3.

ℓ	k	n	solutions in H	CNF clauses	LRAT additions
8	9	390	45478	90957	77148
9	10	533	140250	280501	180907
10	11	707	535635	1071271	682374
8	8	365	47825	95651	47743
9	9	505	184325	368651	288821
10	10	676	705419	1410839	946883

Table 2. Single-value certificates (Lemma 10), each proving $S_2(k;\ell) \le n$. Top: the remaining cases $k = \ell + 1$ of $\ell = 8, 9, 10$, with $n = F(k,\ell)$. Bottom: the diagonal values $WS_8(2)$, $WS_9(2)$, $WS_{10}(2)$, with $n = \frac{k^3+4k^2-5k+2}{2}$.

Each certificate for $\ell \le 7$ was subjected to four complementary verification methods.

(1) *Independent checker* (not importing the search code): exact checks of (i) and (ii), re-derivation of the CNF and comparison with the stored file, and numerical spot checks of every edge for $k_0 \le k \le k_0 + 200$ and $k = 10^3, 10^4, 10^6$.
(2) *LRAT*: the unsatisfiability proofs produced by CaDiCaL [7] were accepted by `lrat-check` [9] and by the verified checker `cake_lpr` [8].
(3) *Second, independently written implementation*: a separate implementation, written and run independently of ours, re-checked (i) and (ii), rebuilt each CNF exactly, and replayed each LRAT proof by unit propagation along the hints to the empty clause. All five certificates passed. A further implementation from the same independent source was also run on the full repository, including SHA-256 checks of the original files, and again all five passed.
(4) *Lean 4*: see §5.

The cases $\ell = 8, 9, 10$. The symbolic certificates for $(\ell, k_0) = (8, 10)$, $(9, 11)$ and $(10, 12)$ passed checks (1) and (2). The second implementation from (3) was also run on them, with all checks passing, but by us rather than by its author, so this is not an independent replay. For the single-value certificates of Table 2, each member of H was checked by a script that does not share code with the search (exactly k summands, exactly ℓ distinct summand values, correct sum, all integers in $[1, n]$), the CNF was rebuilt from H, and the LRAT proofs were accepted by `lrat-check` and `cake_lpr`. As a sanity check, for the three rows with $k = \ell + 1$ the same search finds a good colouring for $n - 1$ (Gaiser's three-block colouring), confirmed by a separate checker. For the three diagonal rows, the colouring of Proposition 4 of $[1, n - 1]$ was confirmed by two separate checkers (for $k = 8, 9$ it is also what the search finds). An independent replay and a Lean formalisation of the cases $\ell = 8, 9, 10$ are pending.
Three further certificates of the same kind, with target $F(k, \ell) + E$ on $k \geq \ell$ for $\ell = 5, 6, 7$, give independent re-proofs of the known values $WS_5(2) = 101$, $WS_6(2) = 166$ and $WS_7(2) = 253$. They passed the same checks (1)–(3).

4.1. **The case** $\ell = 11$**.** Theorem 3(d) at $k = 12$ is the single-value certificate $S_2(12; 11) \leq 915 = F(12, 11)$ (Lemma 10; about 2.05 million hyperedges; files in `evidence/oq64_l11/k12`). For $k \geq 13$ one symbolic certificate in the sense of Section 3 was out of reach, so the colourings were split into cubes. Fix the colour of 1 by symmetry. A *prefix* is a binary string P; its cube consists of the colourings in which the integer $j + 2$ has the colour of 1 if $P_j = 0$ and the other colour if $P_j = 1$ $(0 \leq j < |P|)$. We use a family $\mathcal{P}$ of 222 prefixes of lengths 6 to 9 (Table 3) that is prefix-free and satisfies $\sum_{P \in \mathcal{P}} 2^{-|P|} = 1$, so the cubes partition all colourings. For each $P \in \mathcal{P}$ there is a symbolic certificate with $k_0 = 13$ (conditions (i) and (ii) of Section 3, checked for every $k \geq 13$) whose CNF, together with unit clauses for 1 and for the integers fixed by P, is unsatisfiable. Hence no colouring of $[1, F(k, 11)]$ is good for any $k \geq 13$, i.e. $S_2(k; 11) \leq F(k, 11)$; Theorem 2 gives equality. Finally $S_2(11; 11) = WS_{11}(2) \geq 881 > 836 = F(11, 11)$ by Proposition 4, so $K(11) = 12$.

prefix length	cubes	weight $\sum 2^{9-\lvert P\rvert}$
6	7	56
7	43	172
8	112	224
9	60	60
total	222	$512 = 2^9$

TABLE 3. The prefix partition for $\ell = 11$, $k \geq 13$. In all, the 222 certificates have 2,333,853 hyperedges (one cube has at most 22,638 hyperedges and 886 vertices) and their LRAT proofs 4,619,781 clause additions.

How the cubes were found and checked. The cube search ran on the project owner's computer with the project's search program `symeng_v16` (`evidence/oq64_l8`) under a cube-splitting scheduler; neither is part of the trusted base, only the hyperedge lists they returned are used. From these lists we built the 222 certificates and checked each with the checker of (1) above (exact symbolic conditions for all $k \geq 13$, CNF re-derivation, numerical spot checks for $13 \leq k \leq 60$ and $k = 10^3, 10^4, 10^6$), a separate check that every unit clause is implied by the prefix, and `lrat-check` and `cake_lpr` on the CaDiCaL proofs. A second, independently written implementation, written and run by a second party (GPT, see Appendix A), re-checked the partition and the archive hashes, all 2,333,853 hyperedges with exact arithmetic for every $k \geq 13$, the unit clauses, rebuilt every CNF, and replayed all 222 LRAT proofs by unit propagation along the hints to the empty clause. All checks passed. That replay did not include the $k = 12$ certificate. The replay commands are `verify_all.sh l11k13` and `verify_all.sh l11k13ind`.

5. Lean formalisation

The statement is formalised in Lean 4 (v4.35.0, core library only) as follows (ASCII rendering of the source).

```
def ExactSol (k l N : Nat) (c : Nat -> Bool) : Prop :=
  exists (x : Nat -> Nat) (vals : List Nat),
    (forall i, i <= k -> 1 <= x i /\ x i <= N) /\
    ((List.range k).map x).sum = x k /\
    vals.Nodup /\ vals.length = l + 1 /\
    (forall i, i <= k -> x i in vals) /\
    (forall v in vals, exists i, i <= k /\ x i = v) /\
    (forall i, i <= k -> c (x i) = c (x k))

def FormulaHolds (k l : Nat) : Prop :=
  forall M, (forall c : Nat -> Bool, ExactSol k l M c) <-> F k l <= M
```

Theorems `K3`, …, `K7` (the case $\ell = 8$ is not formalised) state that `FormulaHolds k l` for all k from the threshold on and, for $\ell = 5, 6, 7$, that it fails at $k = \ell$. The formalisation contains:

- a generic proof of Lemma 8, in which conditions (i) and (ii) are decided by computation;
- an LRAT checker with a soundness proof;
- Gaiser's lower bound for all $2 \le \ell \le k$;
- Proposition 4 for all $\ell \ge 5$, which formalises the construction of [1].
- Theorem 5 for all $k \ge \ell \ge 2$ and all j (module `OQ64.Four`), with the instances $S_2(25; 24) > F(25, 24)$, $S_2(26; 25) > F(26, 25)$ and $S_2(22; 22) \ge 6242$.

The only axioms used are `propext`, `Classical.choice` and `Quot.sound`, and `leanchecker` accepts every module. The formalisation has so far been built and checked only within this project; an independent replay is pending.

6. Uniqueness of the extremal colouring

Call Gaiser's three-block colouring of $[1, F - 1]$ (red $[1, s - 1]$, blue $[s, t - 1]$, red $[t, F - 1]$ with $s = k + D$, $t = ks + D$) the *extremal colouring*.

Proposition 11 (computer-assisted)**.** *For each of the twelve pairs*

$$(k, \ell) = (3, 3), (4, 3), (5, 4), (6, 5), (7, 5), (7, 6), (8, 6), (8, 7), (9, 7), (9, 8), (10, 9), (11, 10),$$

the extremal colouring is, up to swapping the colours, the only good colouring of $[1, F(k, \ell) - 1]$.

Each case is certified by a list H of solutions as in Lemma 10 together with one extra clause excluding the extremal colouring (with 1 red); the resulting CNF is unsatisfiable, with LRAT proofs accepted by `lrat-check` and `cake_lpr`. The diagonal case behaves differently: for $k = \ell = 4$ there are three good colourings of $[1, 51]$.

Conjecture 12. *For $\ell \ge 3$ and $k \ge \max(\ell + 1, K(\ell))$, the extremal colouring is the only good colouring of $[1, F(k, \ell) - 1]$ up to swapping the colours.*

The search trees behind Table 2 have a simple shape: for $k = \ell + 1$, the search effectively enumerates the colourings of $[2, \approx 2\ell - 2]$, and each of them is refuted by a short chain of forced colours (typically 5–10 solutions). The number of leaves grows like $2^{2\ell - 3}$. Any conceptual proof of $K(\ell) = \ell + 1$ for a range of ℓ would have to organise these short chains into finitely many patterns, and by Corollary 6 such a proof can only work for $\ell \le 23$.

7. Partial results for general ℓ

Added in v1.0. In v1.2 a second party (GPT) reviewed Proposition 15, Lemma 16 and Lemma 13 as part of the review of Section 8; it also wrote out the infinite-parameter inequalities behind Lemma 16 in full and checked them. The rest of this section has not been checked by a second party. Throughout, $\ell \geq 3$, $k = \ell + e$ with $e \geq 1$, $N = F(k, \ell)$, and χ is a good colouring of $[1, N]$ with $\chi(1)$ red. For a set S of ℓ distinct integers of one colour, any $\sum_{v \in S} \mu(v) v$ with $\mu \geq 1$ and $\sum \mu = k$ is a *k-sum* of S; if it is at most N, it has the other colour.

Lemma 13 (k-sums of an interval)**.** *Let $\ell \geq 3$ and let I be an interval of $L + 1 \geq \ell$ consecutive integers starting at a. The k-sums of the ℓ-subsets of I are exactly the integers in $[ka + D, k(a + L) - D]$, except that for $e = 1$ and $|I| = \ell + 1$ the single value ΣI is missing.*

Sketch. For consecutive ℓ-blocks the k-sums form intervals of length $e(\ell - 1) + 1$ which, for $e \geq 2$, overlap; for $e = 1$ consecutive blocks leave single gaps, each filled by the set $[c, c+\ell+1] \setminus \{c+2, c+\ell\}$ with the extra copy on $c + 1$ (or by its reflection), as long as $|I| \geq \ell + 2$. □

Proposition 14. *If $\ell \geq 5$, the smallest blue integer is at most $\ell + 1$.*

Sketch. If $[1, \ell + 1]$ is red, Lemma 13 shows that the red k-sums cover a blue block starting at $s = k + D$ (for $e = 1$ the block $[s, s + 2\ell - 1]$ misses only $s + \ell$), whose blue k-sums contain $t = ks + D$ and F; so t and F are red. The red solution $1, 1, 2, \ldots, \ell - 1, t$ with $e - 1$ further copies of 1 has sum $t + k - 1 + E = F$, a contradiction. For $e = 1$, F is reached inside the block while avoiding $s + \ell$ by an explicit family of sets. □

Proposition 15 (balanced windows)**.** *Let $\ell \geq 6$. If both colours have at least $\ell - 1$ elements in $[1, 2\ell]$, then χ restricted to $[1, 2\ell]$ is the parity colouring (odd red, even blue).*

Sketch. If a colour C has ℓ elements and the other colour C' has $\ell - 1$ elements in $[1, 2\ell]$, and some $b \in C'$ satisfies $b + d = y + r'$ with $y, d, r' \in C$, then five explicit identities (valid for every $e \geq 1$) produce two different solutions with the same sum and different forced colours; all values are at most N when $\ell \geq 6$. If no such b exists for either choice of C, then both colour classes are closed under $x + y - z$ inside $[1, 2\ell]$, which forces the parity colouring. □

For the parity colouring, an explicit collision exists when k is even (one level of forcing). For k odd we use three levels.

Lemma 16 (parity window, k odd)**.** *Let $\ell \geq 6$, $e \geq 1$ and $k = \ell + e$ odd. No good colouring of $[1, F(k, \ell)]$ is the parity colouring on $[1, 2\ell]$.*

Proof sketch. Let O and E be the odd and even numbers in $[1, 2\ell]$ (red and blue). The red k-sums of O with extra summands from O are the numbers $\ell^2 + \sigma$, where σ runs over the $e(\ell - 1) + 1$ values $e, e + 2, \ldots, e(2\ell - 1)$; call this set A. It is forced blue. Take $m = 1$ for $\ell \in \{6, 7\}$ and $m = \lceil \ell/2 - \sqrt{\ell - 1} \rceil$ otherwise. Let G (resp. G') be the m largest (resp. smallest) even numbers of E, and let $L, L' \subseteq A$ have m elements each. The blue k-sum M of $(E \setminus G) \cup L$ with the extra copies on 2 is forced red. The red k-sum w of $(O \setminus \{2\ell - 1\}) \cup \{M\}$ with the extra copies on 1 is forced blue. The blue k-sum w' of $(E \setminus G') \cup L'$ with the extra copies on 2ℓ is forced red. Then $w = w'$ is equivalent to $\Sigma L' - \Sigma L = r$ with

$$r = (\ell - 1)^2 - e(2\ell - 3) - 2m(\ell - m),$$

which is even because k is odd. Since m-subsets of an arithmetic progression of step 2 realise every even difference of sums up to $2m(e(\ell - 1) + 1 - m)$, it remains to check $|r| \leq 2m(e(\ell - 1) + 1 - m)$ and $w \leq F(k, \ell)$. Writing $\ell = s^2 + 1$ and $m = \ell/2 - s + d$ with $0 \leq d < 1$, both become polynomial inequalities in s, d, e. At $e = 1$ they reduce to $-4d(2s - d) \leq 0$ and $-(s^2 - 1)^2 \leq 0$, and both sides are monotone in e for $\ell \geq 8$. The size condition becomes $F - w \geq s^4(s - 1) + 2s^3 - 3s^2 - s - 2 + e(\cdots) > 0$.

The cases $\ell = 6, 7$ with $m = 1$ are linear in e. The construction was also checked explicitly for $6 \le \ell \le 400$ and $e \le 120$. □

Hence *every hypothetical good colouring of* $[1, F(k,\ell)]$ *with* $\ell \ge 6$ *has a colour with at most* $\ell - 2$ *elements in* $[1, 2\ell]$. Section 8 handles these sparse windows for $e \ge (\frac{\sqrt{2}-1}{2})\ell + 4$. Below that range the following was observed: computationally, for $\ell \le 11$ and $k \ge \ell + 2$ every colouring of $[1, 2\ell]$ is refuted by propagation alone, but for larger ℓ, and for $k = \ell + 1$, sparse windows need further case distinctions. For large ℓ this gap cannot be closed: the colourings of Theorem 5 have exactly such a sparse window (blue $= \{2, \dots, j+1\}$ with $j \approx \ell/4$), and for $\ell \ge 24$ they are good on $[1, F(\ell+1, \ell)]$.

A forcing lemma for sparse windows. The four-block construction of Theorem 5 suggests isolating the small blue interval in the initial window. The following elementary forcing lemma supplies a sufficient condition under which that window cannot extend to a good colouring. It is not an optimality theorem for four-block colourings.

Lemma 17 (Two-stage forcing)**.** *Let* $\ell \ge 11$, $e \ge 1$, $k = \ell + e$, $1 \le j \le \ell - 1$, *and* $n = F(k,\ell)$. *Suppose the restriction of a good red–blue colouring of* $[1, n]$ *to* $[1, 2\ell]$ *is red on* $\{1\} \cup [j+2, 2\ell]$ *and blue on* $[2, j+1]$. *Write* $q = \ell - j$ *and*

$$A_e = e + 1 + \frac{(\ell-1)(2j+\ell+2)}{2}, \qquad L = (\ell-1)q,$$
$$V_e = \frac{j(j+3)}{2} + qA_e + \frac{q(q-1)}{2} + 2e, \quad H = q((\ell-2)q+1),$$
$$P_e = e + 1 + \frac{(\ell-2)(2j+\ell+1)}{2}.$$

If $P_e \le H$ *and* $V_e + P_e \le n$, *no such good colouring exists.*

Proof. We use that the sums of r distinct members of a consecutive integer interval $[a, a+M]$ fill the entire interval from $ra + \binom{r}{2}$ to $ra + \binom{r}{2} + r(M - r + 1)$. Indeed, each integer excess $h = rt + u$, $0 \le u < r$, is realised by shifting all r consecutive choices by t and shifting the last u of them by one more step.

Choose $\ell - 1$ distinct red summands in $[j+2, 2\ell]$ and take 1 with multiplicity $e + 1$. Their valid k-sums run through $[A_e, A_e + L]$. The bound $A_e + L \le n$ holds for every $\ell \ge 11$ and $e \ge 1$, so this entire interval is forced blue.

Now choose all j small blue integers $2, \dots, j+1$, together with q distinct blue integers from $[A_e, A_e + L]$, and repeat 2 another e times. The resulting valid k-sums fill $[V_e, V_e + H]$. Every sum not exceeding n is forced red. Thus V_e and $V_e + P_e$ are red under the stated inequalities.

Finally, take as red summands V_e, the $\ell - 2$ distinct integers $j+2, \dots, j+\ell-1$, and 1 with multiplicity $e + 1$. They have exactly ℓ distinct values, their sum is $V_e + P_e$, and their right-hand side has the same red colour. This contradicts the assumption of a good colouring. □

The parameter V_e is exactly the lower-bound endpoint $V(k,\ell,j)$ from Theorem 5; specifically its m equals $A_e - 1$. Thus $V_e > F$ yields an explicit good four-block colouring, while the stronger opposite-side inequalities $P_e \le H$ and $F - V_e \ge P_e$ exclude extensions of the corresponding initial window. The region between these sufficient conditions remains open.

For example, when $(\ell, j, e) = (23, 10, 1)$, one obtains $k = 24$, $F = 7155$, $A_e = 497$, $L = 286$, $V_e = 6606$, $H = 3562$ and $P_e = 464$. The forced-blue interval is $[497, 783]$, and $7070 = 6606 + 464 \le 7155$ is forced red. Yet 7070 is also the red sum of 6606, $12, 13, \dots, 32$, and two copies of 1, a contradiction.

At $e = 1$, the criterion applies to 100 of the 208 pairs $11 \le \ell \le 23$, $1 \le j \le \ell - 1$ (a finite exact-integer enumeration), and if it holds at $e = 1$ it holds for all $1 \le e \le 1 + H - P_1$. This is a count of the lemma's sufficient conditions, not an exhaustive proof for all sparse windows. In particular, the difficult small-j cases have $V_e + P_e > F$ and require additional colour information.

The same three steps work when the blue part B of $[2, 2\ell]$ is an arbitrary set of $j \le \ell - 1$ elements: in the first step one takes r summands from a red run $[a, a+t-1] \subseteq [2, 2\ell]$ together with $\ell - 1 - r$ further red *anchors*, in the second step one repeats $\min B$ instead of 2, and in the third step one uses the $\ell - 1$ smallest red elements. For example, for $\ell = 11$, $k = 12$ and $B = \{2, 4, 6\}$ (run $[7, 22]$, $r = 8$, anchors $\{3, 5\}$) the forced blue interval is $[94, 158]$, the forced red interval starts at 794, and $794 + 80 = 874 \le 915 = F(12, 11)$ gives a contradiction. For $\ell = 11, 12$ this simple criterion excludes only about 1.5% of all blue sets, so it is far from a treatment of general sparse windows. Computationally, for $11 \le \ell \le 23$ and $k = \ell + 1, \dots, \ell + 5$ every window W_j is excluded by an ordinary hypergraph certificate in the sense of Lemma 10 (1040 certificates, each accepted by `lrat-check` and `cake_lpr`; for $k \ge \ell + 3$ unit propagation from the window alone already gives a contradiction). The lemma remains valid, with the weaker hypothesis $P_e \le H + e(j-1)$, if the e extra blue summands are spread over $[2, j+1]$ instead of being placed on 2, because sums of e elements of $[2, j+1]$ fill $[2e, (j+1)e]$. Combining this with further certificates for the finitely many remaining pairs ($6 \le e \le 31$; 1470 certificates in all, each accepted by `lrat-check` and `cake_lpr`), with Proposition 15 for $j = \ell - 1$, and with an explicit form of Gaiser's argument for $k \ge (3\ell^2 - 3\ell + 2)/2$, we find: *for $11 \le \ell \le 23$ and every $k \ge \ell + 1$, no good colouring of $[1, F(k, \ell)]$ has a window W_j.* For $\ell \ge 24$ the four-block colourings of Theorem 5 have exactly such a window. We also note an obstruction: in the windows W_j with $\ell \le 23$ not covered by the lemma one has $V_e + P_e > F$, and then any red k-sum that uses a summand $\ge V_e$ exceeds F, so further progress must come from integers below A_e.

8. Asymptotics of $K(\ell)$

New in v1.2. The results of this section are a computer-assisted proof candidate and are not peer reviewed. The arguments are written out in full below, and the exact symbolic and finite checks they rely on are in `evidence/oq64_asymptotic`. A second party (GPT) reviewed the whole section, wrote its own checkers and ran them; this review included Proposition 15 and Lemma 16, on which the upper bound depends (Appendix A). Throughout, $a_* = \frac{\sqrt{2}-1}{2} \approx 0.20711$.

Theorem 18. *(a) For every $\ell \ge 50$,*

$$(1 + a_*)\,\ell - 4 \;<\; K(\ell) \;<\; (1 + a_*)\,\ell + 5.$$

In particular $K(\ell) = \frac{1+\sqrt{2}}{2}\,\ell + O(1)$.

(b) For $6 \le \ell \le 49$, $K(\ell) < (1 + a_*)\ell + 9$.

(c) $S_2(20; 19) \ge 4181 > 4163 = F(20, 19)$ and $S_2(21; 20) \ge 4861 > 4812 = F(21, 20)$. Hence $K(19) \ge 21$ and $K(20) \ge 22$.

The lower bounds come from a three-block colouring (Proposition 19). It improves Theorem 5, whose constant is $\frac{9}{8}$. The upper bound uses a new forcing template (Lemma 22) for windows with a sparse colour, together with Proposition 15, Lemma 16, Lemma 13 and a short chain argument for the remaining windows.

Lower bound: three blocks.

Proposition 19. *Let $k \ge \ell \ge 2$, $e = k - \ell$, $1 \le j \le \ell - 1$ and $q = \ell - j$. Put*

$$m = \frac{\ell(\ell+1)}{2} + \ell j + e(j+1) - 1, \qquad V_3 = \frac{j(j+1)}{2} + q(m+1) + \binom{q}{2} + e.$$

The colouring of $[1, V_3 - 1]$ with red class $[1, j] \cup [m+1, V_3 - 1]$ and blue class $[j+1, m]$ is good for $(k, \ell, V_3 - 1)$. Hence $S_2(k; \ell) \ge V_3$.

Proof. Every k-sum with exactly ℓ distinct summand values exceeds each of its summands. A blue k-sum is at least $\sum_{i=1}^{\ell}(j+i) + e(j+1) = m + 1$, so it is not blue. A red k-sum uses at most j

distinct values from $[1, j]$. Hence at least q distinct values are $\geq m+1$, and its sum is at least $\sum_{i=1}^{j} i + \sum_{i=1}^{q}(m+i) + e = V_3$. So it is not red either. □

Compared with Theorem 5, the smallest blue value is now $j+1$ instead of 2. So the e extra copies in the least blue sum contribute $e(j+1)$, which enters V_3 multiplied by q; this changes the leading constant. The same proof shows that the colouring avoids all solutions with *at least* ℓ distinct summand values (the relaxed variant of Open problem 4), because for each colour the least sum only grows with the number of distinct values.

Corollary 20. *(a) Let $\ell \geq 50$. For every integer e with $1 \leq e \leq a_*\ell - 4$ there is a j with $V_3 > F(\ell+e, \ell)$. Hence $K(\ell) > (1+a_*)\ell - 4$.*

(b) With $j = 5$: $V_3 = 4181$ for $(k, \ell) = (20, 19)$ and $V_3 = 4861$ for $(21, 20)$.

(c) On the diagonal $e = 0$, $\max_j V_3$ exceeds the bound of Proposition 4 for every $k \geq 20$; for example $WS_{20}(2) = S_2(20; 20) \geq 4770 > 4751$. The leading term is still $\frac{9}{16}k^3$.

Proof. (a) Expanding, $G(j) = V_3 - F(\ell+e, \ell) = -Aj^2 + \beta j + \gamma$ with $A = e + \ell - 1 > 0$. The real maximum is $\gamma + \beta^2/(4A)$, attained at $j_* = \frac{(\ell-1)(2e+\ell-2)}{4(e+\ell-1)}$. An integer j within $\frac{1}{2}$ of j_* loses at most $A/4$. So some integer j satisfies $G(j) \geq 1$ as soon as $H = 4A\gamma + \beta^2 - A^2 - 4A \geq 0$. Substitute $\ell = 50 + u$ and $e = a_*\ell - 4$. Then H is a polynomial in u whose coefficients lie in $\mathbb{Q}(\sqrt{2})$, and each coefficient is ≥ 0; the same substitution shows $1 \leq j_* \leq \ell - 1$. For fixed j, $\partial G/\partial e = q(j+1) + 1 - \partial F/\partial e \leq -\frac{\ell^2}{4} - \ell + \frac{1}{4} - 2e < 0$, so the same j works for every smaller $e \geq 1$. With $j = x\ell$ and $e = \varepsilon\ell$, the cubic part of G is $-\frac{\ell^3}{2}\left[\varepsilon(2x^2 - 2x + 1) + 2x^2 - x\right]$. Some x makes it positive iff $\varepsilon < \max_x \frac{x - 2x^2}{2x^2 - 2x + 1} = a_*$, with the maximum at $x = \frac{2-\sqrt{2}}{2}$; so the constant a_* is the threshold of this family. (b) is a direct evaluation. To prove (c) for every $k \geq 20$, write $k = 4n + r$ with $0 \leq r \leq 3$ and choose $j = n = \lfloor k/4 \rfloor$. If $A(k) = (k^3 + 4k^2 - 5k + 2)/2$ is the Ahmed–Boza–Revuelta–Sanz lower bound, exact expansion gives

$$V_3(k, 0, n) - A(k) = \begin{cases} (4n-1)(n^2 - 5n + 1), & r = 0, \\ 2n^2(2n - 9), & r = 1, \\ (4n+1)(n^2 - 4n - 1), & r = 2, \\ (n-4)(2n+1)^2, & r = 3. \end{cases}$$

All four expressions are positive for $n \geq 5$, covering $k \geq 20$; at $k = 19$ the difference is zero. For $19 \leq \ell \leq 49$ the largest admissible e was computed exactly for each ℓ (`finite_table.py`); the bound of Theorem 18(a) also holds there. □

The colourings in (b) and (c) were also confirmed to be good by two separately written exact bit-set dynamic programmes, one by Claude and one by GPT. Each checker was first validated on Gaiser's colouring, which is good on $[1, F-1]$ and cannot be extended. A further run confirmed (b) for the relaxed variant.

Upper bound. Let $\ell \geq 6$, $e \geq 1$, $N = F(\ell+e, \ell)$, and let χ be a good colouring of $[1, N]$ with $\chi(1)$ red. Put $W = [1, 2\ell]$. One of the following holds.

(B) Both colours have at least $\ell - 1$ elements in W. This is impossible by Proposition 15, Lemma 16 and the one-step collision for the parity colouring when k is even (§7).

(R) W is monochromatic, hence red.

(U) One colour D has j elements in W with $1 \leq j \leq \ell - 2$; the other colour M has $\ell + q \geq \ell + 2$ elements in W, where $q = \ell - j$.

Lemma 21 (all-red window)**.** *Case (R) is impossible for every $e \geq 1$.*

Proof. By Lemma 13 applied to $[1, 2\ell]$, every integer in $I = [k + D, 2k\ell - D]$ is a red k-sum. These integers satisfy $2\ell < k + D$ and $2k\ell - D < N$, so they are blue. The interval I has at least $\ell + 2$ elements, so by Lemma 13 again the blue k-sums of I cover $[k(k+D)+D,\ k(2k\ell-D)-D] \supseteq [t, N]$, where $t = k(k+D)+D$. In particular t is red. The red multiset $1^{(k-\ell+1)}, 2, 3, \ldots, \ell-1, t$ has exactly ℓ distinct values and sum $t + k - 1 + E = N$, which is red. This is a monochromatic solution. □

(Lemma 13 needs $|I| \geq \ell + 2$ only when $e = 1$. In that case the gap between consecutive blocks $[c, c + \ell - 1]$ is filled by $[c, c + \ell - 2] \cup \{c + \ell + 1\}$ with the extra copy on $c + \ell - 2$, or, for the last gap, by $\{c - 1\} \cup [c + 2, c + \ell]$ with the extra copy on $c + 2$.)

Lemma 22 (T1-fill)**.** *In case (U) with $2 \leq j \leq \ell - 2$, let $S = \{m_1 < \cdots < m_\ell\}$ be the ℓ least elements of $M \cap W$. There are $y, r', d \in S$ and $b \in D \cap W$ with $b = y + r' - d$; here $y = r'$ is allowed. Put*

$$R' = S \setminus \{y\}, \quad u = \min R', \quad r = \min(S \setminus \{r'\}), \quad \Sigma_0 = \sum S + (e-1)u,$$

let $z_1, \ldots, z_{q-1}$ be the $q - 1$ least elements of $S \setminus \{r, r'\}$, and let $d_b = \min(D \cap W \setminus \{b\})$. Then

$$w = \sum R' + \sum(D \cap W) + q\,\Sigma_0 + \textstyle\sum_i z_i + r + d + e\,d_b + (e-1)u$$

satisfies $w > N$.

Proof. The triple exists. Suppose first that D meets $[1, m_\ell]$ and that no such triple exists. Then S is closed under $(y, r', d) \mapsto y + r' - d$ whenever the result lies in $[1, m_\ell]$. Let $\delta = m_2 - m_1$. The integer $2m_1 - m_2$ cannot be a positive element of S, so $\delta \geq m_1$. Moving up by δ and down by δ shows that $S = (m_1 + \delta\mathbb{Z}) \cap [m_1, m_\ell]$. If $\delta = 1$ then $m_1 = 1$ and $D \cap [1, m_\ell] = \emptyset$, a contradiction. If $\delta \geq 2$ then $m_\ell \geq m_1 + 2(\ell - 1) > \ell + j \geq m_\ell$, again a contradiction; the last inequality holds because at most j elements of $[1, m_\ell]$ are not in M. If instead $D \cap [1, m_\ell] = \emptyset$, then $S = [1, \ell]$, and some $b \in D$ satisfies $\ell + 1 \leq b \leq 2\ell - 1$ (because $j \geq 2$); take $y = \ell$, $r' = b + 1 - \ell$, $d = 1$. In every case $d \notin \{y, r'\}$, because $b \notin M$.

The forcing chain. The values $L_1 = \Sigma_0 + r$, $L_2 = \Sigma_0 + r'$ and $Z_i = \Sigma_0 + z_i$ are k-sums of S: each element of S once, u a further $e - 1$ times, and one more copy of r, r' or z_i. So they are forced to D if they are at most N. They are pairwise distinct (since $r \neq r'$ and $z_i \notin \{r, r'\}$) and larger than 2ℓ. Put $B_3 = (D \cap W \setminus \{b\}) \cup \{Z_1, \ldots, Z_{q-1}\}$, a set of $(j - 1) + (q - 1) = \ell - 2$ values of colour D. Then

$$\begin{aligned}
M_v &= \textstyle\sum B_3 + b + L_1 + e\,d_b && \text{is a } k\text{-sum of the } D\text{-set } B_3 \cup \{b, L_1\},\ \text{forced to } M;\\
w &= \textstyle\sum R' + M_v + (e-1)u + d && \text{is a } k\text{-sum of the } M\text{-set } R' \cup \{M_v\},\ \text{forced to } D;\\
w' &= \textstyle\sum B_3 + L_1 + L_2 + e\,d_b && \text{is a } k\text{-sum of the } D\text{-set } B_3 \cup \{L_1, L_2\},\ \text{forced to } M.
\end{aligned}$$

Each of these sets has exactly ℓ distinct values. Moreover $w - w' = b + d - y - r' = 0$, because B_3 cancels. So the value $w = w'$ would have both colours. Every value in the chain is at most w, so $w \leq N$ is impossible. Expanding M_v gives the displayed formula for w. □

For $q = 1$ (no values Z_i) this is the template behind Proposition 15. The new point is that the missing minority values can be replaced by forced values, whose contributions cancel in $w - w'$.

Lemma 23. *If $\ell \geq 50$ and $e \geq a_*\ell + 4$, every window of case (U) with $2 \leq j \leq \ell - 2$ violates Lemma 22: there $w \leq N$, a contradiction. For $6 \leq \ell \leq 49$ the same holds for every $e \geq e_U(\ell)$, where $e_U(\ell) \leq 14$ is the value listed in `finite_table.py`.*

Proof. Because at most j elements of $[1, m_i]$ lie outside M, we have $m_i \leq i + j$. This gives $\sum R' \leq \sum S \leq \ell j + \frac{\ell(\ell+1)}{2}$, $z_i \leq j + i + 2$, $r \leq m_2 \leq j + 2$ and $d \leq \ell + j$. Also $\sum(D \cap W) \leq \sum_{i=1}^{j}(2\ell - j + i)$ and $d_b \leq 2\ell$. If M is red (case I), then $m_1 = 1$, and $y \neq 1$ can be arranged, because $y = r' = 1$ would give $b \leq 0$; hence $u = 1$. If M is blue (case II), $u \leq m_2 \leq j + 2$. Since w is increasing in each of these quantities, $N - w \geq P(j)$ for an explicit quadratic $P(j) = \alpha j^2 + \beta j + \gamma$, with $\alpha = \ell + 1$

in case I and $\alpha = \ell + e$ in case II. We show $4\alpha\gamma - \beta^2 \geq 0$, which gives $P(j) \geq 0$ for every real j. Substitute $\ell = 50 + u'$ and $e = a_*\ell + 4$. In both cases $4\alpha\gamma - \beta^2$ becomes a polynomial in u' whose coefficients lie in $\mathbb{Q}(\sqrt{2})$, and every coefficient is nonnegative. Finally, $\partial P/\partial e \geq \frac{\ell^2}{4} - 2\ell + 2e - \frac{5}{4} > 0$ for $\ell \geq 9$, and in case I always; so every larger e works as well. With the additive constant 3 in place of 4 the case II check fails (negative control). For $6 \leq \ell \leq 49$ the inequalities $P(j) \geq 0$ were evaluated exactly for every integer $2 \leq j \leq \ell - 2$ at the tabulated threshold $e_U(\ell)$. To justify all larger e (not merely the finite scan), note that

$$\frac{\partial P}{\partial e} = \begin{cases} 2e + \frac{\ell^2 - \ell + 2}{2} - (\ell - j + 1), & \text{case I,} \\ 2e + \frac{\ell^2 - \ell + 2}{2} - (\ell - j + 1)(j + 2), & \text{case II.} \end{cases}$$

The first is positive for $\ell \geq 6$, $e \geq 1$. In case II, $(\ell - j + 1)(j + 2) \leq (\ell + 3)^2/4$, so the derivative is at least $2e + (\ell^2 - 8\ell - 5)/4$, which is positive for $\ell \geq 9$, $e \geq 1$, and also for $\ell \in \{6, 7, 8\}$, $e \geq 3$. The exact table has $e_U(6) = 9$, $e_U(7) = 10$, $e_U(8) = 10$, so monotonicity establishes the asserted bound for all $e \geq e_U(\ell)$. □

Lemma 24 (one minority element)**.** *Case (U) with $j = 1$ is impossible for $\ell \geq 7$, $e \geq 3$, and for $\ell = 6$, $e \geq 4$.*

Proof. If M is blue, then $D \cap W = \{1\}$ and $M \supseteq [2, 2\ell]$. Take $S = [2, \ell + 1]$; then its e-fold sums fill $[2e, (\ell + 1)e]$. If M is red, let $D \cap W = \{b_0\}$, and take $S = [1, \ell]$ if $b_0 > \ell$, and $S = [1, \ell + 1] \setminus \{b_0\}$ otherwise. In the second case $S - 1$ has gaps at most 2 and contains two consecutive integers starting at an offset $a \leq 2$. In general, let $O \subseteq [0, \Delta]$ contain 0 and Δ, have all gaps at most r, and contain $[a, a + r - 1]$. Then $(e - 1)O \supseteq \bigcup_{i \leq e-2}(i\Delta + O)$ has gaps at most r, so $eO \supseteq [a, (e - 1)\Delta + a + r - 1]$. In both red cases this gives $eS \supseteq [e + 2, e + (e - 1)(\ell - 1)]$.

So $\sum S + eS$ contains an interval $[A, A + L]$ of values forced to D. Its $\ell - 1$-element subsets, together with the window value of D and e extra copies of it, give k-sums filling an interval of length $(\ell - 1)(L + 2 - \ell)$ beginning at some V_0; these are forced to M. Adding the $\ell - 1$ least elements of M and e copies of m_1 to V_0 gives a k-sum $V_0 + P$ of M, which lands back in that interval. The four required inequalities (all values at most N, $L + 1 \geq \ell - 1$, $P \leq (\ell - 1)(L + 2 - \ell)$ and $V_0 + P \leq N$) are polynomial in ℓ, e. Substituting $\ell = 7 + u'$, $e = 3 + s$ (respectively $\ell = 6 + u'$, $e = 4 + s$), each has only nonnegative coefficients. □

Proof of Theorem 18. The lower bounds are Corollary 20. For the upper bounds, the cases (B) and (R) are impossible for every $e \geq 1$; case (U) with $j = 1$ is impossible for $e \geq 4$ (Lemma 24); and with $2 \leq j \leq \ell - 2$ it is impossible once $e \geq a_*\ell + 4$ for $\ell \geq 50$, or $e \geq e_U(\ell)$ for $6 \leq \ell \leq 49$ (Lemma 23). Hence $S_2(k; \ell) \leq F(k, \ell)$ for every $k \geq \ell + \lceil a_*\ell + 4 \rceil$. Together with Theorem 2 this gives equality, and $K(\ell) \leq \ell + \lceil a_*\ell + 4 \rceil < (1 + a_*)\ell + 5$. For $6 \leq \ell \leq 49$ the exact table gives $K(\ell) \leq \ell + \max(e_U(\ell), 4)$, and the largest value of $\ell + \max(e_U(\ell), 4) - (1 + a_*)\ell$ in that range is $8.5502\ldots < 9$. □

How the checks were made. All nonnegativity statements above are exact: `sympy` computations with $\sqrt{2}$ kept symbolic, together with exact integer evaluation. The following were also checked.

- Lemma 22: explicit hyperedges were built and verified on random windows.
- The minimum of w over all admissible parameter choices was computed for 6807 windows of colourings known to be good (the three-block colourings above and those of Theorem 5). In each case it exceeds N, as it must.
- Every window with $2 \leq j \leq \ell - 2$ for $\ell = 6, 7, 8$ was checked exhaustively.

These checks were run with two separately written sets of programs (Appendix A). The constants -4 and $+5$ are what this method gives. For the three-block windows themselves, the exact minimum of w already becomes $\leq N$ about three or four steps of e beyond the last good colouring, so the true value of $K(\ell)$ lies within a few units of $(1 + a_*)\ell$. Theorem 29 below makes this precise.

8.1. **Exact thresholds for $12 \le \ell \le 20$ (new in v1.3).** *New in v1.3.* The results of this subsection and the next are computer-assisted proof candidates and are not peer reviewed. The method and all computations are first-party (Claude); the state of the second-party review is stated in the status box and in Appendix A. The data, the programs and all 14,723 LRAT proofs are in `evidence/oq64_exact`.

Theorem 25 (computer-assisted)**.** *(a) $K(\ell) = \ell + 1$ for every $12 \le \ell \le 18$. Hence the least ℓ with $S_2(\ell+1;\ell) > F(\ell+1,\ell)$ is $\ell = 19$.*
(b) $K(19) = 21$ and $K(20) = 22$.
(c) $S_2(20;19) = 4181$ and $S_2(21;20) = 4861$.

By (a), $S_2(k;\ell) = F(k,\ell)$ for all $k > \ell$ when $\ell \le 18$; so $(k,\ell) = (20,19)$ is the smallest pair with $k > \ell$ and $S_2(k;\ell) > F(k,\ell)$, and (c) gives its value. In both cases of (c), $S_2 = \max\{F, \max_j V_3\}$, the form asked about in Open problem 2. In (b), the lower bounds of Theorem 18(c) are attained.

An exact form of Lemma 22. Call a tuple $(y, r', d) \in S^3$ *admissible* if $b := y + r' - d \in D \cap W$; then automatically $d \notin \{y, r'\}$, because $b \notin M$. The forcing chain in the proof of Lemma 22 uses nothing about the tuple except admissibility. So the conclusion $w > N$ holds for *every* admissible tuple, with u, r, z_i, d_b determined by the tuple as in the lemma. Write $w^*(W)$ for the least value of w over all admissible tuples; then a window with $w^*(W) \le N$ is impossible. Using $\sum R' = \sum S - y$, $d - y = r' - b$, and the fact that $\{r\} \cup \{z_i\}$ is the set of the q least elements of $S \setminus \{r'\}$, the formula of Lemma 22 becomes

$$w = (q+1)\textstyle\sum S + \sum(D \cap W) + (q+1)(e-1)\,u + \big(r' + Q(r')\big) - b + e\,d_b, \tag{1}$$

where $Q(r')$ is the sum of the q least elements of $S \setminus \{r'\}$. Here $u = m_1$ unless $y = m_1$; $r' + Q(r')$ equals the sum $\Sigma_{q+1}(S)$ of the $q+1$ least elements of S when r' is one of them, and exceeds it otherwise; and $d_b = \min(D \cap W)$ unless b is that minimum.

Lemma 26 (reduction)**.** *In case (U) with $2 \le j \le \ell - 2$, let $m_\ell = \max S$ and let W' be obtained from W by recolouring every position $> m_\ell$ with the colour M. If $D(W') = D \cap [1, m_\ell]$ has at least two elements, then $w^*(W) \le w^*(W') - t(\sum S - 2\ell) \le w^*(W')$, where $t = |D \cap W| - |D(W')|$.*

Proof. W' is in case (U) with the same colours M, D and the same set S, and $2 \le |D(W')| \le j$. Every admissible tuple of W' is admissible for W, since $D(W') \subseteq D \cap W$. For such a tuple, u and $r' + Q(r')$ are computed from S alone, except that q drops by t when passing from W' to W; d_b is the same, because the two least elements of D lie in $D(W')$. In (1), passing from W' to W adds the t new minority values (each at most 2ℓ) to $\sum D$, lowers q by t in the two terms $(q+1)\sum S$ and $(q+1)(e-1)u$, and replaces $Q(r')$ by the sum of its $q - t$ least terms. Hence $w(W) \le w(W') + 2\ell t - t\big(\sum S + (e-1)u\big) \le w(W') - t(\sum S - 2\ell)$. Finally $\sum S \ge \ell(\ell+1)/2 \ge 2\ell$ for $\ell \ge 3$. □

Lemma 27 (monotonicity in e)**.** *Let $\ell \ge 9$. For a fixed window and a fixed admissible tuple, $w - F(\ell + e, \ell)$ is strictly decreasing in e. In particular, if a window has $w^*(W) \le F(\ell + e_0, \ell)$ for $k = \ell + e_0$, then it has $w^*(W) \le F(\ell + e, \ell)$ for every $e \ge e_0$.*

Proof. By (1), increasing e by one increases w by $(q+1)u + d_b \le (\ell - j + 1)(j+2) + 2\ell \le (\ell+3)^2/4 + 2\ell$, using $u \le m_2 \le j + 2$. It increases $F(\ell + e, \ell)$ by $2(\ell + e) + 1 + \frac{(\ell+1)(\ell-2)}{2} + 2$. The difference is at least $\frac{\ell^2 - 8\ell - 1}{4} + 2e > 0$. □

Proof of Theorem 25. Lower bounds. For (a), $K(\ell) \ge \ell + 1$ by Proposition 4. For (b) and (c) they are Theorem 18(c).

Upper bounds. Fix ℓ and $k = \ell + e$. Let $N = F(k,\ell)$, or $N = 4181$ and $N = 4861$ in (c), with $e = 1$. Let χ be a good colouring of $[1, N]$ with $\chi(1)$ red. Case (B) is impossible, and so is (R) (Lemma 21); these arguments produce a monochromatic solution inside $[1, F(k,\ell)] \subseteq [1, N]$. Case

(U) with $j = 1$ is impossible for $e \geq 3$ (Lemma 24). Case (U) with $2 \leq j \leq \ell - 2$ is impossible whenever $w^*(W) \leq N$.

All windows with $2 \leq j \leq \ell - 2$ and $w^*(W) > N$ (the *survivors*) were computed exactly as follows. By Lemma 26, it suffices to do three things:

- enumerate the *reduced* windows (all positions above m_ℓ in M, $|D| \geq 2$);
- examine individually every window over a reduced window that survives;
- examine individually the windows with $|D \cap [1, m_\ell]| \leq 1$, for which $S \subseteq [1, \ell + 1]$.

By Lemma 27, one pass over the reduced windows treats a whole range of e. A reduced window is evaluated at the next e only while it survives. The enumeration is `tools/t1red.c` and `tools/t1multi.c`. For $\ell = 17, 18, 19, 20$ it visits $5.66 \cdot 10^8$, $2.20 \cdot 10^9$, $8.60 \cdot 10^9$ and $3.36 \cdot 10^{10}$ reduced windows.

The results are as follows.

- For every ℓ in the theorem there are no survivors at $e = e_P(\ell)$, where $e_P = 3$ for $12 \leq \ell \leq 15$ and $e_P = 4$ for $16 \leq \ell \leq 20$. By Lemma 27 there are then no survivors at any $e \geq e_P$.
- Each survivor at $e < e_P$, and each of the 2ℓ windows with $j = 1$ at $e \leq 2$, was refuted by a *per-window certificate*. A search program recorded monochromatic solutions (hyperedges) on $[1, N]$. Each hyperedge was re-checked from the definition. The colouring CNF on $[1, N]$, with the window fixed by unit clauses, was proved unsatisfiable by CaDiCaL, and the LRAT proof was accepted by both `lrat-check` and `cake_lpr`.
- For (b) only $e \geq 2$ is needed, and for (c) only $e = 1$ with the stated N.

Table 4 lists the counts. Together with Theorem 2, this gives $S_2(k; \ell) = F(k, \ell)$ for all $k \geq \ell + 1$ in (a), for all $k \geq \ell + 2$ in (b), and $S_2 \leq N$ in (c). □

ℓ	e_P	survivors at $e = 1, 2, 3$	per-window certificates	result
12	3	66, 2, 0	90, 26	$K = 13$
13	3	121, 5, 0	147, 31	$K = 14$
14	3	209, 9, 0	237, 37	$K = 15$
15	3	389, 17, 0	419, 47	$K = 16$
16	4	716, 33, 1	748, 65, 1	$K = 17$
17	4	1311, 63, 3	1345, 97, 3	$K = 18$
18	4	2425, 127, 4	2461, 163, 4	$K = 19$
19	4	–, 236, 8	–, 274, 8	$K = 21$
20	4	–, 481, 15	–, 521, 15	$K = 22$
19		$e = 1$, $N = 4181$: 3520	3558	$S_2(20; 19) = 4181$
20		$e = 1$, $N = 4861$: 4386	4426	$S_2(21; 20) = 4861$

TABLE 4. The computation behind Theorem 25. Certificate counts include the 2ℓ windows with $j = 1$ at $e \leq 2$. There are 14,723 certificates in all.

Checks.

- For $\ell = 12, 13$ the survivor sets coincide with those of a brute-force run over all $2^{2\ell-1}$ windows.
- For $9 \leq \ell \leq 11$ and $e \leq 3$, a second implementation recomputed w^* for every window with $2 \leq j \leq \ell - 2$: `t1fill_witness.py` from the v1.2 evidence. It agrees with the C code on every window, and for every window it calls refuted it builds the explicit hyperedges of Lemma 22 and checks that the chain closes.
- The one-pass and segmented runs reproduce the per-e survivor files byte for byte.

- The hardest certificates are those of the prefix windows; at $\ell = 18$, $e = 1$ the three-block colouring falls short of F by only 2. The largest LRAT proof, for red $= [1, 4]$ at $N = 4861$, has 83,357 hyperedges.

8.2. **Prefix dominance (new in v1.3).** For $2 \le j \le \ell - 2$ let P_j be the window with red part $[1, j]$ and blue part $[j + 1, 2\ell]$. It is the window of the three-block colouring of Proposition 19; it lies in case (U) with M blue, $S = [j + 1, j + \ell]$ and $D \cap W = [1, j]$. In (1) each term is bounded below by its value at the tuple $(y, r', d) = (j+2, j+1, j+3)$, for which $b = j$, $u = j+1$ and $d_b = 1$. Hence

$$w^*(P_j) = (q+1)\Big(\ell j + \tfrac{\ell(\ell+1)}{2}\Big) + \tfrac{j(j+1)}{2} + (q+1)(e-1)(j+1) + (q+1)j + \tfrac{(q+1)(q+2)}{2} - j + e, \tag{2}$$

$$G_P(\ell, e, j) := w^*(P_j) - F(\ell + e, \ell) = -e^2 - \tfrac{e(\ell^2+\ell-2)}{2} - (e + \ell - 1)j^2 + \Big(e\ell + \tfrac{\ell^2-\ell-2}{2}\Big)j + 2\ell, \tag{3}$$

and $w^*(P_j) - V_3 = e(j + 1) + j(\ell - 2) + \ell(\ell + 3)/2$.

Lemma 28 (prefix dominance)**.** *Let $\ell \ge 4$ and $1 \le e \le 2\ell^2 - 4\ell$. Every window W in case (U) with $2 \le j \le \ell - 2$ satisfies $w^*(W) \le w^*(P_{j'})$ for some $2 \le j' \le j$. Consequently the largest value of $w^*(W) - N$ over all such windows is $\max_{2\le j'\le \ell-2} G_P(\ell, e, j') - (N - F)$.*

Proof. Call an admissible tuple *good* if $y \ne m_1$ and r' is one of the $q + 1$ least elements of S. For a good tuple, (1) reads $w = (q + 1)\sum S + \sum D + (q + 1)(e - 1)m_1 + \Sigma_{q+1}(S) - b + e\, d_b$ (we write D for $D \cap W$).

Reduced windows. Let $|D \cap [1, m_\ell]| \ge 2$. By Lemma 26 we may assume that W is reduced, so $S \sqcup D = [1, \ell + j]$ and $D_j < m_\ell = \ell + j$, where $D = \{D_1 < \cdots < D_j\}$. Then $\sum S + \sum D = \sum S_{P_j} + \sum D_{P_j}$. Put $\delta = \sum_i (D_i - i) = \sum S_{P_j} - \sum S \ge 0$ and $\Delta = \Sigma_{q+1}(S_{P_j}) - \Sigma_{q+1}(S) \ge 0$; also $m_i \le i + j$. For a good tuple,

$$w - w^*(P_j) = -q\delta - (q + 1)(e - 1)(j + 1 - m_1) - \Delta + (j - b) + e(d_b - 1). \tag{4}$$

Case II ($1 \in D$). Let p be maximal with $[1, p] \subseteq D$, so $m_1 = p + 1$. The sequence $D_i - i$ is non-decreasing. Let $i^* = \max\{i : D_i - i \le q + p - 1\} \ge p$; then $D_i - i \ge q + p$ for $i > i^*$, so $\delta \ge (j - i^*)(q + p)$. We choose a good tuple with $b \ge \max(2, i^*)$, so $d_b = 1$ and $j - b \le j - i^* \le q\delta$, and (4) gives $w \le w^*(P_j)$.

- If $i^* > p$, take $b = D_{i^*}$, $r' = m_1$, and for $y \in Y = S \cap (b, \ell + j]$ put $d = y - (b - m_1)$. There are $|Y| = \ell - D_{i^*} + i^* \ge j - p + 1$ distinct values d. They lie in $(m_1, \ell + j]$, where only $j - p$ integers are not in S. So some $d \in S$, and the tuple is good.
- If $i^* = p$ and $2 \le p < j$, then $D_{p+1} \ge q + 2p + 1 \ge p + 4$; take $(y, r', d) = (p + 2, p + 1, p + 3)$, so $b = p$.
- If $p = j$, then $W = P_j$.
- If $i^* = p = 1 < j$, then $D_2 \ge q + 3$ and $[2, q + 2] \subseteq S$; take $(y, r', d) = (D_2 - q, q + 2, 2)$, so $b = D_2$ and $r' = m_{q+1}$.

Case I ($1 \in S$, so $m_1 = 1$). Let $t = D_1 - 1 \ge 1$. Since $m_i = i$ for $i \le t$, we have $\delta \ge tj$ and $\Delta \ge j$. Also $(q + 1)j - (\ell + j - 1) = (j - 1)(q - 1) \ge 0$, so $(q + 1)j \ge \ell + j - 1 \ge \max(t, D_2)$.

- If $t \ge 2$, take $(y, r', d) = (D_j + 1, 1, 2)$: $b = D_j \ge t + j$ and $d_b = t + 1$. By (4), $w - w^*(P_j) \le -qtj - (e - 1)t - t + et = -qtj < 0$.
- If $t = 1$ (so $2 \in D$), the $\ell - 1$ elements of $S \setminus \{1\} \subseteq [3, \ell + j]$ contain two consecutive integers $x, x + 1$; otherwise D would need at least $\ell - 2$ separating elements besides 2, so $j \ge \ell - 1$. Take $(y, r', d) = (x + 1, 1, x)$: $b = 2 = D_1$ and $d_b = D_2$. Now $m_i = i + 1$ for $2 \le i \le D_2 - 2$, so $\delta \ge j + (D_2 - 3)(j - 1)$. By (4), $w - w^*(P_j) \le -q\big(j + (D_2 - 3)(j - 1)\big) - (e - 1)(D_2 - 1) - j + (j - 2) + e(D_2 - 1) = -qj - (D_2 - 3)\big(q(j - 1) - 1\big) < 0$.

Low windows ($|D \cap [1, m_\ell]| \le 1$). Write $D'' = D \cap (m_\ell, 2\ell]$ and $\beta = \min D''$. We use (1) directly. In cases (L2) and (L1) we choose an admissible tuple with $b = \beta$; in (L0a) we take $b = D_2$, and in (L0b) we take $b = D_1 = \beta$. Each tuple has $y \neq m_1$, so $u = m_1$, but it need not be good: r' need not be among the $q+1$ least elements of S. The excess of $r' + Q(r')$ over $\Sigma_{q+1}(S)$ is bounded explicitly in each case. We then bound the full difference with $w^*(P_j)$.

(L2) Case II, $S = [2, \ell+1]$, $D = \{1\} \cup D''$ with $D'' \subseteq [\ell+2, 2\ell]$. Take $(y, r', d) = (\ell+1, \beta-\ell+1, 2)$: $u = 2$, $d_b = 1$, and $r' + Q(r') \le \Sigma_{q+1}(S) + (j-1)$. With $\sum S - \sum S_{P_j} = -\ell(j-1)$, $\sum D \le 1 + 2\ell(j-1)$ and $\Sigma_{q+1}(S) - \Sigma_{q+1}(S_{P_j}) = -(q+1)(j-1)$:

$$w - w^*(P_j) \le -(q-1)\ell(j-1) - q(j-1) - \tfrac{j(j+1)}{2} + j - \ell - 1 < 0.$$

(L1) Case I, $S = [1, \ell+1] \setminus \{a\}$ with $2 \le a \le \ell$, $D = \{a\} \cup D''$ with $D'' \subseteq [\ell+2, 2\ell]$. Take $(\ell+1, \beta-\ell, 1)$ if $\beta - \ell \neq a$; otherwise take $(\ell+1, a+1, 2)$ if $a \neq 2$, and $(\ell+1, 4, 3)$ if $a = 2$. Then $b = \beta$, $u = 1$ and $d_b = a$, and $r' + Q(r') \le \Sigma_{q+1}(S) + j$. With $\sum S - \sum S_{P_j} = -(\ell(j-1) + a - 1)$, $\sum D \le a + 2\ell(j-1)$, $\Sigma_{q+1}(S) - \Sigma_{q+1}(S_{P_j}) \le -(q+1)(j-1)$ and $(q+1)j \ge a - 1$:

$$w - w^*(P_j) \le -(q-1)\ell(j-1) - (q-1)(a-1) - \tfrac{j(j+1)}{2} - (q-1)j - j < 0.$$

(L0a) Case I, $S = [1, \ell]$, $D \subseteq [\ell+1, 2\ell]$, $D_2 \le 2\ell - 1$. Take $(\ell, D_2 + 1 - \ell, 1)$: $b = D_2$, $u = 1$, $d_b = D_1 \le 2\ell - j + 1$, and $r' + Q(r') \le \Sigma_{q+1}(S) + (j-1)$. Using $(q+1)j - (2\ell - j) = (j-2)(\ell - j) \ge 0$, i.e. $(q+1)j \ge D_1 - 1$:

$$w - w^*(P_j) \le -(q-1)\ell j - \tfrac{j(j+1)}{2} - (q+1)j < 0.$$

(L0b) Case I, $S = [1, \ell]$, $D = \{D_1, 2\ell\}$ (so $j = 2$, $q = \ell - 2$). Take $(\ell, D_1 + 1 - \ell, 1)$: $b = D_1$, $u = 1$, $d_b = 2\ell$, and $r' + Q(r') \le \Sigma_{\ell-1}(S) + 1$. Direct expansion gives

$$w - w^*(P_2) \le e - 2\ell^2 + 4\ell.$$

This is the only case in which e survives the estimate: the term $e(d_b - 1) = e(2\ell - 1)$ is offset only by $2(\ell - 1)(e - 1)$. It is ≤ 0 precisely for $e \le 2\ell^2 - 4\ell$, the hypothesis of the lemma.

In all cases $w^*(W) \le w \le w^*(P_{j'})$, with $j' = |D \cap [1, m_\ell]|$ for reduced windows and $j' = j$ for low windows. □

The constructions in this proof were checked by machine on every window with $2 \le j \le \ell - 2$, for $6 \le \ell \le 10$ and several values of e up to $2\ell^2 - 4\ell$. Beyond that bound the check fails exactly at (L0b), as predicted. The conclusion of the lemma was also checked against the exact minimum of w in two ways. For $10 \le \ell \le 13$ and several e, the check covered all reduced and low windows (`lemmaP/t1dom.c`). For $\ell = 10, 11$ and $e = 1, 3, 6$, the second party checked every window with its own program. Across all enumerations of Theorem 25, the largest surviving value of $w^* - N$ equals the prefix value at every (ℓ, e): for example 72 at $(\ell, e) = (19, 3)$, 238 and 106 at $(20, 2), (20, 3)$, 300 for $N = 4181$ and 326 for $N = 4861$.

Theorem 29. *Let $\ell \ge 9$ and $e_P(\ell) = \min\{e \ge 1 : G_P(\ell, e, j) \le 0$ for all $2 \le j \le \ell - 2\}$. Then $K(\ell) \le \ell + \max\{3, e_P(\ell)\}$. For every $\ell \ge 19$,*

$$(1 + a_*)\ell - 3.63 \;<\; K(\ell) \;\le\; (1 + a_*)\ell + 16 - 11\sqrt{2} \;<\; (1 + a_*)\ell + 0.444.$$

The lower bound does not use Lemma 28.

Proof. The first assertion. Let $e_0 = \max\{3, e_P(\ell)\}$. Cases (B) and (R) are impossible for every e, and case (U) with $j = 1$ for $e \ge 3$. For $2 \le j \le \ell - 2$, Lemma 28 at $e = e_0$ gives $w^*(W) \le F$; this needs $e_0 \le 2\ell^2 - 4\ell$, which holds since $e_P(\ell) \le \lceil a_*\ell \rceil$ (below) and $e_P(\ell) \le 4$ for $9 \le \ell \le 18$. By Lemma 27 the same tuple refutes the window for every $e \ge e_0$. With Theorem 2 this gives $S_2(k; \ell) = F(k, \ell)$ for $k \ge \ell + e_0$.

The upper bound for $\ell \geq 19$. Write $G_P = -Aj^2 + \beta j + \gamma$ with $A = e + \ell - 1 > 0$. Its maximum over real j is negative if and only if $\beta^2 + 4A\gamma < 0$, and $\partial G_P/\partial e < 0$.

- For $\ell \geq 55$, substitute $e = a_*\ell + c_0 - 1$, with $c_0 = 16 - 11\sqrt{2}$, and $\ell = 55 + u$. Then $\beta^2 + 4A\gamma$ becomes a cubic in u whose four coefficients, in $\mathbb{Q}(\sqrt{2})$, are all negative. Hence $e_P(\ell) \leq \lceil a_*\ell + c_0 - 1 \rceil < a_*\ell + c_0$.
- For $19 \leq \ell \leq 54$, exact evaluation gives $e_P(\ell) - a_*\ell \leq c_0$. Equality holds at $\ell = 22$, where $e_P = 5$ and $5 - 22a_* = 16 - 11\sqrt{2}$.
- In both ranges $e_P \geq 4 > 3$.

(The same substitution with $c_0 = 1$ already works from $\ell = 19$; this gives the bound $e_P(\ell) \leq \lceil a_*\ell \rceil$ used above.)

The lower bound. Use the criterion of Corollary 20(a). Substitute $e = a_*\ell - 3.63$ and $\ell = 19 + u$. Then $H = 4A\gamma + \beta^2 - A^2 - 4A$ has only positive coefficients, and the vertex j_* satisfies $\frac{3}{2} \leq j_* \leq \ell - \frac{3}{2}$. So some integer j has $V_3 - F \geq 1$ at $e = \lfloor a_*\ell - 3.63 \rfloor$, and $K(\ell) \geq \ell + \lfloor a_*\ell - 3.63 \rfloor + 1 > (1 + a_*)\ell - 3.63$. □

Remarks.

- By Theorem 29, $K(\ell)$ lies in the integer interval $[\ell + e_L(\ell) + 1,\ \ell + \max\{3, e_P(\ell)\}]$, where $e_L(\ell)$ is the largest e with $\max_j V_3 > F(\ell + e, \ell)$. Numerically $e_P - e_L \in \{3, 4\}$ for $19 \leq \ell \leq 20000$. For real parameters the gap tends to $\frac{3}{2} + \sqrt{2} \approx 2.914$; this is a heuristic, not a statement about the integer sequence.
- The constant $16 - 11\sqrt{2} \approx 0.4437$ is best possible for this method: the difference $e_P(\ell) - a_*\ell$ attains it at $\ell = 22$, where $e_P(22) = 5$. This does not assert that $K(22)$ attains the upper bound. For the lower constant, the infimum of $\ell + e_L(\ell) + 1 - (1 + a_*)\ell$ over $19 \leq \ell \leq 20000$ is -3.6199; numerically this is close to $-\frac{3}{2}(1 + \sqrt{2}) \approx -3.6213$.
- Inequality (4) also yields a safe pruning rule for enumerating reduced windows, found by the second-party reviewer. Suppose the first r elements of D are fixed and D_r is the largest of them. Then $\delta \geq \delta_{\text{partial}} + (j - r)(D_r - r)$, and the whole branch is refuted as soon as

$$q\big(\delta_{\text{partial}} + (j - r)(D_r - r)\big) \geq G_P + (j - 1) + [\text{case I}]\, e(2\ell - 1).$$

  In the reviewer's prototype this cuts $3.36 \cdot 10^{10}$ reduced windows at $\ell = 20$, $e = 2$ to about $2 \cdot 10^4$ search nodes. It reproduces every survivor set of Table 4, window by window. It is not used in the proof of Theorem 25.

9. Three colours: Open Question 6.2

For $r \geq 1$ let $S_r(k; \ell)$ be the least n such that every r-colouring of $[1, n]$ contains a monochromatic solution of $x_1 + \cdots + x_k = x_{k+1}$ with exactly $\ell + 1$ distinct integers. For $\ell = 2$ such a solution consists of i copies of u and $k - i$ copies of v ($u \neq v$, $1 \leq i \leq k - 1$) with sum w. Gaiser proved $S_2(k; 2) = k^2 + 2k$ for $k \geq 3$ and $S_3(k; 2) \geq k^3 + 3k^2 + k - 1$, and asked whether equality holds for all large k [11, Open Question 6.2].

Theorem 30 (computer-assisted)**.** *For every $k \geq 3$, $S_3(k; 2) = k^3 + 3k^2 + 2k - 2$. In particular the answer to Open Question 6.2 is negative.*

Lower bound. Let $T = k^3 + 2k^2$ and $N = k^3 + 3k^2 + 2k - 3$. Colour $[1, N]$ by

$$R = [1, k] \cup [k^2 + k + 1, k^2 + 2k - 1] \cup [T + 1, T + k - 1] \cup [T + k^2 + k - 1, N],$$
$$B = [k + 1, k^2 + k] \cup [T + k, T + k^2 + k - 2], \qquad G = [k^2 + 2k, T].$$

This is Gaiser's colouring with an extra red block of $k - 1$ integers at the top. For a solution $w = iu + (k - i)v$ with $u < v$ one has $ku + (v - u) \leq w \leq kv - (v - u)$, and Table 5 shows that no class contains a solution. The statement is formalised in Lean 4 for all $k \geq 2$ (theorem

`Gaiser.lower`). *This lower bound was first recorded publicly by the trureturing project of the Omega Institute on 5 October 2026* [10]*; we found it independently and make no priority claim for it.*

The block check. Name the blocks in increasing order $R_1 = [1, k]$, $B_1 = [k+1, k^2+k]$, $R_2 = [k^2+k+1, k^2+2k-1]$, $G_1 = [k^2+2k, T]$, $R_3 = [T+1, T+k-1]$, $B_2 = [T+k, T+k^2+k-2]$, $R_4 = [T+k^2+k-1, N]$. If $u < v$ lie in one block $[a, b]$ then $ka+1 \le w \le kb-1$; if $u \in [a, b]$ and $v \in [c, d]$ with $b < c$, then $(k-1)a + c \le w \le b + (k-1)d$. Table 5 lists all fourteen same-colour pairs of blocks. Only the pair R_1, R_3 needs more: if $v \in R_3$ occurs at least twice then $w \ge 2(T+1) + k - 2 > N$; otherwise $w = (k-1)u + v \in [T+k, T+k^2-1] \subseteq B_2$. All inequalities in the table hold for every $k \ge 3$.

Pair of blocks	w lies in	Bound
R_1, R_1	B_1	$k+1 \le w \le k^2-1$
R_1, R_2	G_1	$k^2+2k \le w \le k^3+k^2-2k+1$
R_1, R_3	B_2 or $w > N$	see the text
R_1, R_4	$w > N$	$w \ge N+1$
R_2, R_2	G_1	$k^3+k^2+k+1 \le w \le T-k-1$
R_2, R_3	$w > N$	$w \ge 2k^3+2k^2$
R_2, R_4	$w > N$	$w \ge 2k^3+3k^2+k-2$
R_3, R_3; R_3, R_4; R_4, R_4	$w > N$	$w \ge kT+k+1$
B_1, B_1	$R_2 \cup G_1$	$k^2+k+1 \le w \le k^3+k^2-1$
B_1, B_2	R_4 or $w > N$	$w \ge T+k^2+k-1$
B_2, B_2	$w > N$	$w \ge k(T+k)+1$
G_1, G_1	$R_3 \cup B_2 \cup R_4$ or $w > N$	$w \ge T+1$

TABLE 5. The fourteen same-colour pairs of blocks in the lower-bound colouring for Theorem 30, with $u < v$. In every row w lies in a block of another colour or beyond N.

Upper bound. We use only the solutions $w = (k-1)u + v$ with $u \ne v$. Let χ be a 3-colouring of $[1, F_3]$, $F_3 = k^3+3k^2+2k-2$, with no such monochromatic triple, $\chi(1)$ red, and let a be the largest j with $[1, j]$ red. Then $a \le k$, since $(k-1)\cdot 1 + 2 = k+1$, and $a+1$ may be taken blue. For $k \ge 8$ the seven cases $a = 1, 2, 3$, $a = k-2, k-1, k$ and $4 \le a \le k-3$ are each refuted by a symbolic certificate in the sense of Section 3: its vertices are integer polynomials in k (or in k and a for $4 \le a \le k-3$), its edges are polynomial identities $w = (k-1)u + v$, and some vertices are known to be red (they lie in $[1, a]$) or blue (the vertex $a+1$). The range, sign and fact conditions are certified on the whole case region by substituting $k = 8+t$ (and $a = 4+s$, $k = 7+s+t$) and checking that all coefficients are nonnegative. Each three-colour CNF is unsatisfiable. For $3 \le k \le 7$ the value is settled by direct SAT runs at $n = F_3$.

Verification. All LRAT proofs (seven case certificates and the five direct runs for $3 \le k \le 7$) were accepted by `lrat-check` and `cake_lpr`. A separately written checker re-proved all symbolic conditions and rebuilt every CNF; a second, independently written implementation re-checked the semantics and replayed all twelve LRAT proofs (7/7 and 5/5). The whole theorem is also formalised in Lean 4, core library only (`RSN.S3_two_eq`: for $k \ge 3$ and all M, every 3-colouring of $[1, M]$ has a monochromatic solution if and only if $F_3 \le M$); it uses only the axioms `propext`, `Classical.choice` and `Quot.sound`, and `leanchecker` accepts all 107 modules. The Lean development has so far been built only within this project.

Remark 31. Writing $S_r(k)$ for the classical r-colour generalized Schur numbers ($S_1(k) = k$, $S_2(k) = k^2+k-1$, $S_3(k) = k^3+2k^2-2$), both $S_2(k; 2) = 1 + S_1(k) + S_2(k)$ and Theorem 30, $S_3(k; 2) = 1 + S_1(k) + S_2(k) + S_3(k)$, fit the pattern $S_r(k; 2) = 1 + \sum_{j \le r} S_j(k)$. A computation for

case	region	vertices	edges
$a = 1$	$k \geq 8$	54	326
$a = 2$	$k \geq 8$	55	345
$a = 3$	$k \geq 8$	83	561
$a = k$	$k \geq 8$	55	301
$a = k - 1$	$k \geq 8$	50	276
$a = k - 2$	$k \geq 8$	46	265
$4 \leq a \leq k - 3$	$k \geq 8$	86	579

TABLE 6. Case certificates for the upper bound in Theorem 30.

$r = 4$, $k = 3$ also agrees (229), but its unsatisfiability proof was checked and then not archived, so we record it only as evidence.

10. Open problems

1. *Exact thresholds beyond $\ell = 20$.* (The question of v1.2, whether $K(\ell) = \ell+1$ for $12 \leq \ell \leq 18$, is answered by Theorem 25.) Let $e_L(\ell)$ be the largest e with $\max_j V_3 > F(\ell + e, \ell)$. Is $K(\ell) = \ell + e_L(\ell) + 1$ for every $\ell \geq 19$, that is, is the three-block family always exactly optimal for the threshold? This holds for $\ell = 19, 20$ (Theorem 25(b)). By Theorem 29, $K(\ell)$ lies in $[\ell + e_L(\ell) + 1, \ell + \max\{3, e_P(\ell)\}]$, an interval of three or four integers for $19 \leq \ell \leq 20000$.
2. *Exact values above F.* Is $S_2(k; \ell) = \max\{F(k, \ell), \max_j V_3\}$ whenever $k > \ell \geq 5$? For $\ell \leq 18$ this reduces to $S_2 = F$, which holds by Theorem 25(a). At the two smallest pairs with $S_2 > F$, namely $(20, 19)$ and $(21, 20)$, it holds by Theorem 25(c). (Version 1.2 asked this for $k \geq \ell$. On the diagonal the answer is negative: for example $WS_8(2) = 365 > 344 = F(8, 8) > 330 = \max_j V_3$.)
3. *Optimality of the constructions.* Among colourings consisting of a small initial set followed by two blocks, a local search found nothing better than the initial interval $[1, j]$ of Proposition 19. Is the three-block colouring extremal among all good colourings when it beats F?
4. *The relaxed variant.* Gaiser also asks about $S_2(k; \tilde{\ell})$, defined via solutions with "at least" a given number of distinct integers. The paper describes it with $c(S) \geq \ell + 1$ on p. 2 but defines it with $c(S) \geq \ell$ on p. 12. Under the second reading, Corollary 6.3 of [11] would fail already for $\ell = 2$, since then $S_2(k; \tilde{2}) = S_2(k) = k^2 + k - 1$. This was first noted in an AI-generated review page [12]. We use the reading $c(S) \geq \ell + 1$, the only one consistent with Corollary 6.3. Under that reading every solution counted by $S_2(k; \ell)$ is also counted by $S_2(k; \tilde{\ell})$, so $S_2(k; \tilde{\ell}) \leq S_2(k; \ell)$, and Gaiser's three-block colouring gives $S_2(k; \tilde{\ell}) \geq F(k, \ell)$. Hence the relaxed threshold is at most $K(\ell)$. The colourings of Theorem 5 and Proposition 19 also avoid all solutions with at least ℓ distinct summand values. So Corollary 6(a),(b), Corollary 20 and Theorem 18 hold for the relaxed threshold as well; in particular it is $\frac{1+\sqrt{2}}{2}\ell + O(1)$. The same applies to Theorems 25 and 29. Their upper bounds transfer because $S_2(k; \tilde{\ell}) \leq S_2(k; \ell)$. Their lower bounds transfer because the colourings used are good for the relaxed variant, and because on the diagonal $k = \ell$ the two notions coincide.

Appendix A. Reproducibility, verification provenance and AI use

Scope. The results are computer-assisted proof candidates, not peer-reviewed proofs. This preprint accompanies the public repository `https://github.com/PYCSON/restricted-schur-proofs` (release v1.3; v1.0 of 28 October 2026, v1.1 of 1 November 2026 and v1.2 of 3 November 2026 are archived unchanged), which contains all certificates, checkers and Lean sources; paths below refer to that repository. The lower bound of Theorem 30 had been recorded publicly before [10]; no priority is claimed for it.

What the certificates establish. Unsatisfiability of a CNF is not by itself the mathematical statement. Each claim also needs: the validity of every hyperedge (polynomial identities and the region conditions on the whole parameter range for symbolic certificates, exact arithmetic for numerical ones), the correctness of the colouring-to-CNF encoding, and, for equalities, the lower-bound colouring. For Theorem 30 these are seven symbolic case certificates ($k \geq 8$) and five direct runs ($3 \leq k \leq 7$). For Theorem 3 they are the symbolic certificates of Table 1 and the single-value certificates of Lemma 10; the diagonal values $WS_8(2), WS_9(2), WS_{10}(2)$ have their own single-value certificates.

Who did what. The search programs, all certificates, most checking scripts, the Lean formalisation and the text of this draft were produced by Claude (Anthropic), working with tools under the project owner's direction; Claude also ran CaDiCaL, `lrat-check`, `cake_lpr` and Lean. A checker called *independently written* below has source code written separately from the search programs; when Claude ran it, that run is *not* a second-party run. GPT (OpenAI) wrote and ran its own checkers: for Theorem 30, a semantic audit of the symbolic conditions, CNF reconstruction and an LRAT replay of all twelve proofs; for Open Question 6.4, a symbolic audit, CNF reconstruction and LRAT replay of eight certificates: the five threshold certificates for $3 \leq \ell \leq 7$ in Table 1 and the three certificates re-proving the known values $WS_5(2), WS_6(2), WS_7(2)$ (42,018 symbolic hyperedges in all; all passed; the two largest JSON files were read in parts, not checked byte for byte). Claude also ran GPT's Python checker on all OQ 6.4 symbolic certificates, including those for $\ell \geq 8$. GPT reviewed Theorem 5, Corollary 6 and the sharpness remark on paper and with a computer algebra system, wrote the block table of Section 9 (re-checked by Claude for $3 \leq k \leq 2000$), and reviewed the Lean statements without running Lean. Codex (OpenAI) ran tests on the project owner's computer; according to its reports, it checked the colouring for $(k, \ell) = (25, 24)$ and replayed the certificate chain for $WS_8(2) = 365$ with an independent encoder, a fresh solver run, its own LRAT checker and negative controls. Its report, logs and outputs are archived with their hashes. For $\ell = 11$, $k \geq 13$ (§4.1, new in v1.1): the cube search was run on the project owner's computer (set up by Codex, restarted by the owner); Claude built the 222 certificates from the returned hyperedge lists and checked them; GPT then wrote and ran its own exact symbolic validator and LRAT checker on all 222 certificates (all passed; source, logs and results are in `evidence/oq64_l11/k13_cubes/independent_review`, where only file paths were adapted for the release), and Claude re-ran GPT's checker on the archived files with the same totals. For Section 8 (new in v1.2), Claude found and wrote out the arguments and the exact checking scripts (`evidence/oq64_asymptotic/scripts`). A separately started Claude instance, not given the drafting history, re-derived the section by hand and wrote its own checking code (`independent_review_claude_subagent`); since it is the same model, this is not counted as a second-party review. GPT then reviewed the section as a second party, including Proposition 15, Lemma 16 and Lemma 13; it wrote and ran its own exact symbolic and finite checkers and bit-set dynamic programmes, and drafted text that was used for this section (`independent_review_gpt`, with its report; passages of the report referring to private correspondence were removed for the release). Claude re-ran GPT's checkers on the release files with the same results. For §8.1 and §8.2 (new in v1.3), Claude designed the exact form of the forcing template, the reduction lemma, the one-pass enumeration and Lemma 28, and wrote and ran all programs. These include the

enumerators `t1red.c` and `t1multi.c`, the per-window certificate pipeline, the regeneration of the stored LRAT proofs, and the replay script `replay_perwin.py`. Claude also wrote a second implementation of the template decision (`fullcheck/`) and ran it on every window for $9 \le \ell \le 11$. GPT reviewed §8.1 and §8.2 as a second party in two rounds. In the first round GPT wrote its own C, Python, SymPy and Z3 programs. With them it re-enumerated all windows for $\ell = 12, 13$ and the reduced windows for $\ell \le 20$, re-checked 4,004,355 hyperedges, and decided 1,009 of the certificate CNFs with Z3, including all 294 for $\ell = 12, 13$. It also found the pruning rule and the constant $16 - 11\sqrt{2}$ of Theorem 29, which Claude re-derived. In the second round GPT reviewed the proofs of Lemma 28 and Theorem 29 line by line, checked every identity and final inequality with SymPy and the lemma against the exact minimum w^* on every window for $\ell = 6, 7, 8$; two wording corrections in the low-window part of Lemma 28 were made at its request. It also confirmed the statements of `OQ64.Three`. For the certificates GPT authored, from a written specification of the data formats and without access to the project's scripts, a Python-standard-library checker (`evidence/oq64_exact/independent_review_gpt/gpt_r9_check.py`) that reconstructs the window and hypergraph clauses, validates every hyperedge, checks the completeness of the window lists, and verifies the pure-RUP LRAT derivations, with a nine-case self-test. Claude executed this checker, unmodified, on the archived certificate collection: 14,723 accepted proofs and 0 failures, self-test 9/9 tamperings rejected; the source SHA-256 is `f177e36c...7af2` (in full in that directory). GPT reviewed the returned logs and checksums. This is an independently written implementation executed by Claude, not an independent execution of the complete data set by GPT. The second party's verdict (8 November 2026) was ACCEPT for Theorem 25, Lemma 28 and Theorem 29 at this level; Lemma 28 was reviewed as a written proof, with symbolic and finite-window machine checks. No second-party replay is claimed for the threshold certificates with $8 \le \ell \le 10$, for the $k = 12$ certificate of $\ell = 11$, for $WS_9(2)$ and $WS_{10}(2)$, or for any Lean development.

Lean. The development `lean/s3k2-full` proves `RSN.S3_two_eq`; `leanchecker` passed 107/107 modules; a strict NanoDa check timed out without a verdict. The development `lean/oq64` contains $K(3), \ldots, K(7)$, Proposition 4 and `OQ64.Four.four_block`; new in v1.3, `OQ64.Three` proves Proposition 19 for all $2 \le \ell \le k$ and all j, and the instances $S_2(20; 19) > F(20, 19)$, $S_2(21; 20) > F(21, 20)$ and $WS_{20}(2) \ge 4770$. These give the lower bounds of Theorem 25(b),(c). A copy of the proof with the bound increased by one fails, as it must, and `leanchecker` accepts the module. Both use Lean v4.35.0 (commit `eb6cafb7290e8bd5f2937522d8d804f58f02bafc`), core library only; the axiom checks show only `propext`, `Classical.choice` and `Quot.sound`. Not formalised: the certificates for $\ell \ge 8$, the diagonal values, the general inequalities of Corollary 6, the upper bounds of §8, and Lemma 28.

Tools. CaDiCaL 3.0.1; `lrat-check` from `drat-trim`; `cake_lpr`. CaDiCaL source commit `c6073042`, `drat-trim` commit `2e3b2dc0`; the SHA-256 of the binaries used are recorded in the repository README. Codex's environment (Windows, Python 3.12.9, CaDiCaL 2.1.2) differs from ours.

Replay. From a fresh clone of the release (commands from the repository READMEs):

```
cd evidence/ub_cert            # Theorem (OQ 6.2), seven symbolic cases
python3 check_kernel.py uni_const1 uni_const2 uni_const3 \
        uni_top0 uni_top1 uni_top2 bi_mid
xz -dc CASE.lrat.xz > /scratch/CASE.lrat     # for each CASE
lrat-check CASE.cnf /scratch/CASE.lrat
cake_lpr CASE.cnf /scratch/CASE.lrat
cd ../oq64_kernel              # Theorem (OQ 6.4), symbolic certificates
python3 check_cert.py cert_3_3.json          # and each other cert_*.json
cd ../../lean/s3k2-full && ulimit -s unlimited && lake build
lake env lean Test/Axioms.lean && lake env leanchecker RSN Gaiser
lake env lean Test/Fault.lean  # 4 tampered checks, all must be rejected
```

```
cd ../oq64 && lake build OQ64.Final OQ64.Final2 OQ64.Four
cd ../.. && ./verify_all.sh asym asymind   # Section 8 (v1.2)
```

A fault-injection test succeeds when it fails with the designated error. The numerical certificates need a file-by-file manifest.

Manifest. The file `MANIFEST.sha256` in the repository lists the SHA-256 of every file; the README maps each claim to its certificate files and replay commands.

Author's contribution. Puyi Sun selected the research questions and directed the collaborative investigation, specified the evidence and reproducibility requirements, coordinated cross-checks between independently developed tools, and made the decisions concerning the scope and release of this preprint. The computational proofs and formal developments reported here were generated and checked with substantial AI assistance, with the provenance and verification status recorded above. The author has not personally replayed every certificate or independently checked every proof step. Accordingly, this release is presented as a computer-assisted proof candidate for external scrutiny, not as a substitute for human mathematical review.

Declaration of generative AI and AI-assisted technologies. During this research and the preparation of the manuscript, the author used Claude (Anthropic), GPT (OpenAI) and Codex (OpenAI) for mathematical exploration, proof and algorithm development, generation and checking of computational certificates, formalisation in Lean, literature assistance and drafting. Some checks were performed by independently developed implementations, but the extent of independent execution varies by result and is detailed above. The author reviewed the research scope and the evidence reports, has disclosed the remaining verification limitations, and takes responsibility for the decision to release this preprint. No AI system is listed as an author.

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